



SECJ2203: Software Engineering

System Documentation (SD)

Koperasi KADA's Automated System

Version 3.0

12 February 2025

Faculty of Computing

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Revision Page

a. Overview

The System Documentation (SD) of the current version of Koperasi KADA's Automated System has the system introduction comprising the scope, purpose, objective and feature to be included in the system. SD provides user personas, user stories and system features. Product backlogs, sprint deliverables, user story with support of sequence and activity diagram. Also covered are non-functional requirements.

b. Target Audience

The target audience are applicants, admin and Ahli Lembaga Koperasi of KADA.

c. Project Team Members

Table 1.1 is the list name of our team members by stating our roles and the status for each assigned task by sections for this SD version (complete, partially complete, incomplete). This table also includes the task and progress status of each team member in this project.

Table 1.1

Member Name	Role	Task	Status
Farra Nurzahin	Project Manager	Project planning	Complete
		Project management	Incomplete
		Create system architecture	Complete
Dayang Farah Farzana	Technical Writer	Gather system and user requirements	Complete
		Create SRS document	Complete
		System documentation	Incomplete
Nawarrah Auni	System Developer	Develop system	Incomplete
		System Testing	Incomplete
		System Implementation	
Safiya Nursyahadah	Designer	System Design	Incomplete
		Design system functional diagram	Complete
		Design draft system	Complete

d. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Farra Nurzahin binti Zaharil Anuar	Completed Topic 1(Introduction), Scope, Section 2.1.2 Section 2.2.1, Section 2.3, Section 2.4.2, Section 2.5	30/11/2024
1.0	Dayang Farah Farzana binti Abang Idham	Completed Overview, Section 1.3, Section 2.1.3, Section 2.2 (System Features), Section 2.3, Section 2.4.3	30/11/2024
1.0	Nawwarah Auni binti Nazrudin	Completed Purpose, Specific Requirements, Section 2.1, Section 2.2.2, Section 2.3, Section 2.4, Section 2.4.1, Section 2.6.1, Section 2.6.4	29/11/2024
1.0	Safiya Nursyahadah binti Masnoor	Completed Target Audience, Section 1.5 (overview), Section 2.1.1, Section 2.2.3 (state diagram), Section 2.3, Section 2.6	30/11/2024
2.0	Farra Nurzahin binti Zaharil Anuar	Correction of 2.1.1, 2.2.1, 2.3, 2.4.1 Completed UC010	12/2/2025
	Dayang Farah Farzana binti Abang Idham	Correction of 2.1.3, 2.2.1, 2.3, Completed UC011	12/2/2025
	Nawwarah Auni binti Nazrudin	Correction of 2.1.2, 2.2.1, 2.2.2, 2.3, 2.4.2 Completed UC009	12/2/2025
	Safiya Nursyahadah binti Masnoor	Correction of 2.1.2, 2.2.3, 2.4.3,	12/2/2025
3.0	Farra Nurzahin binti Zaharil Anuar	Correction of 2.1.1, 2.2, 2.2.1, UC010, 2.4.1	12/2/2025
	Dayang Farah Farzana binti Abang Idham	Correction of 2.1.3, 2.2, 2.2.1, UC011, 2.4.3	12/2/2025
	Nawwarah Auni binti Nazrudin	Correction of 2.1.2, 2.2.1, 2.2.2, UC009, 2.3, 2.4.2	12/2/2025
	Safiya Nursyahadah binti Masnoor	Correction of 2.1.2, 2.2.3, 2.4.2	12/2/2025

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1. Introduction

This part of System Documentation (SD) provides detailed information about the system's development and implementation processes. This document consists of Software Requirements Specification (SRS) to outline the functional and nonfunctional requirements of the system. This to make sure what kind of software in the Koperasi KADA's system must be achieved. The comprehension of Software Design Description (SDD) is also in this document to provide essential detailed design information and architecture for the system's development. Lastly, Software Test Documentation (STD) is included in this document to describe the test plans to validate and verify the software's functionality and performance.

1.1 Purpose

This SD describes the requirements for Koperasi KADA's Automated System. It explains how the system will handle membership and loan management. It also describes the system's features, how users will interact with it, and any rules or limits the system must follow. To help understand the system better, this document uses user stories that describe scenarios to show different users, like staff and members, will interact with the system to complete their tasks.

This document is meant for developers, system administrators, users and stakeholder, Koperasi KADA who need to understand how the system works and how to use it effectively.

1.2 Scope

The software product is the Koperasi KADA automated website system which is going to be built to make managing memberships and processing loans easier for cooperative members. It contains the following main software products including membership management system module, loan management system module, report generator module and notification system module.

Firstly, the membership management system module makes sure the processing of the submission, review and approval of membership applications are secured and also provides the applicants with a real-time update. Secondly, the loan management module eases the submission and approval of loan applications that includes the loan calculator as

well as real-time update. This module only prioritizes loans within the Koperasi KADA and does not support other external loans. Next, the report generator module covers the preparation of detailed reports and statements regarding loans, membership and other activities for the administrators. Lastly, the notification system module is built for sending an automated notification via applicant's email to keep members updated about their application statuses. It only covers the notification from email and not SMS.

This software is built to make managing memberships and processing loans easier for Koperasi KADA. This product ensures the improvement of operational efficiency by automating the manual processes as well as reduces errors and saves time for administrators and ensures data security for Koperasi KADA by having a built-in encryption and access control. The objective of this document is to:

- I. To design a web-based system for KADA's koperasi system.
- II. To develop and test the system to ensure the requirements are met.

1.3 Definitions, Acronyms and Abbreviation

This section provides the definitions, acronyms and abbreviations used in the System Documentation (SD)

- ALK : Ahli Lembaga Koperasi - Board members that responsible for decision-making
- Admin : Staff that responsible for managing and reviewing applications
- Applicant : A user that apply membership or loan through the system
- SD : System Documentation - Detailed document

1.4 References

A. Dennis, B. H. Wixom & D. Tegarden (2015). *System Analysis and Design, An Object-Oriented Approach with UML*, 5th Edition. [E-book]

1.5 Overview

The System Documentation (SD) for Koperasi KADA's Automated System provides a detailed description of the system's purpose, structure, and features. It includes a system introduction that defines its scope, purpose, objectives, and features. This document uses diagrams like use case, class diagram, state machine diagram to visualize the system's

functionality and interactions. Additionally, it covers non-functional requirements and design constraints to ensure the system's quality, performance, and security standard.

The SD is organized into sections for better understanding. It begins with an introduction to the system's goals and requirements, followed by detailed descriptions of personas, system features, and workflows. Specific requirements are explained using user stories supported by diagrams for each use case. The document concludes with sections on performance expectations, design constraints and administrative details like version control and the project team's distributions. This structured approach ensures that the SD is user-friendly and informative for developers, stakeholders and end users.

2. Specific Requirements

This section explains the detailed software requirements necessary for building and testing the system. It provides enough information to guide designers in creating a system that meets these needs and helps testers ensure the system works as intended. Every requirement is described in a way that can be observed or experienced by users, operators, or connected systems. It includes details about all inputs, the action it performs and the outputs.

2.1 Persona

The personas for this system includes applicants, administrators, and members of the Koperasi KADA board (ALK). Each user has specific needs and responsibilities, which are described through their individual stories to provide clear understanding of the user requirements.

2.1.1 Persona 1

User Need

- Applicants need a way to apply for a membership, login into the system, apply for a loan and calculate loan as well as apply for a termination of membership.

User Stories

- As an applicant, I want to apply for a membership and login into the Koperasi Kada Automated System so that I can apply for a loan.
- As a member, I want to fill out a loan application form so that I can provide my details for loan approval.
- As a member, I want to calculate the monthly loan repayment and any applicable dividends so that I can make an informed decision.
- As a member, I want to terminate my membership application so that I can discontinue my association with Koperasi KADA.

When Applicant opens the Koperasi KADA Automated System's website, the interface shows the user's login details. After the Applicant login successfully, the applicant will be taken directly to the main page menu which includes activities such as membership application, loan application, and calculation of loan features.

If the applicant clicks on 'Borang Keanggotaan' under the 'Permohonan Anggota', the applicants are taken to the membership form page from the menu option. The applicant has to complete details in the first part of the form and then press on the 'next' button to go to the next page. On the following page, applicants fill in other parts of the form and click the 'submit' button at the end of the page. Submission shows their membership status where applicants are informed if their application has been reviewed or not.

However, to apply for a loan, an applicant can click on 'Borang Pembiayaan' under the 'Permohonan Pembiayaan'. On this page, applicants will be directed to the loan application page where they can input the amount and type of loan, and the loan duration to get estimated monthly repayments using the loan calculator.

Additionally, in order to proceed further, the applicant has to accept the agreement by clicking on the agreement option. After that, the applicant has to give the details of a guarantor and make sure that guarantor agrees to the terms and conditions. Once an applicant completes the form and submits the application, the system will show the loan application status.

If a member wishes to terminate their membership, they can do so by selecting 'Borang Berhenti' under the 'Permohonan Berhenti' option from the menu. Once the member submitted their application, the system will show the application status.

2.1.2 Persona 2

User Need

Admin need a way to review membership, loan and terminate applications, manage transactions, generate an online statement and report and send automatic notifications to applicants

User Stories

- As an admin, I want to review submitted membership applications so that I can verify the applicant's details.
- As an admin, I want to review and validate loan applications so that I can ensure it aligns with the Koperasi KADA's loan criteria.
- As an admin, I want to review submitted membership termination so that I can verify the main reason for termination.

- As an admin, I want to manage transactions of the members of Koperasi KADA's so that financial records remain accurate and up-to-date for reporting.
- As an admin, I want to generate online statements and reports so that I can provide correct reports' records to the stakeholders.
- As an admin, I want to send automated email notifications to applicants so that they are updated about the status of their applications.
- As an admin, I want to send financial statement to applicants so that they are updated about their financial balance.

Admin has a task on managing the system effectively including reviewing submitted membership, membership termination and loan applications, generating statements and reports as well as sending notifications via email. Admin logs in the system as admin and redirects to the admin page.

If admin chooses 'Membership List' or 'Loan Application List' or 'Membership Termination', they will be redirected to the list of the applicant's name page. Admin click on the not reviewed button to review applicant's details and make sure all details provided are correct and complete.

In both cases, they can click on the 'Complete' if the application is complete and correct and 'Incomplete' if the application is incomplete and not accurate.

If the admin approves a member through 'Approve Member by ALK,' they will be redirected to the 'Status Submission' page to send the application status to the applicant. Approved applicants can then access the 'Financial Statement' page to generate their statement.

The admin can also generate the statement by clicking on 'Generate Statement'. After that, a pop-up of an generated statement will be displayed on the admin screen to confirm the statement. Automated notifications are then sent to the approved applicant through their registered email by clicking on the 'Send' button.

2.1.3 Persona 3

User Need

- ALK needs a way to approve membership and approve loan applications

User Stories

- As an ALK, I want to approve applicants' membership so that I can ensure that only eligible individuals can become membership and earn membership benefits
- As an ALK, I want to approve loan application so that I can let only the eligible applicants to receive loan in line with company's policies
- As an ALK, I want to approve membership termination application so that I can deactivate members' membership

ALK is responsible for membership and loan approval and the decision made is fair inline with company policies. ALK logs in to the system and then is forwarded to a personalized page for ALK.

If ALK decides to review membership applications, then the system will display a pending list for membership applications. The same applies to when ALK chose to review loan applications.

ALK then selects any applications that appear on the pending list of membership and loan approval. ALK evaluates the applications and decides for both membership and loan approval or rejection based on the company's policies. For membership, if the status of the reviewed application is 'Incomplete', then ALK will reject immediately.

In the case of loan approval, ALK is allowed to use the 'Compare' button to compare loan applications with loan terms. The system compares and displays whether or not applicants pass the loan terms. ALK also can directly approve or reject the application without using the button.

ALK is also responsible for terminating membership application approval. Similar to membership and loan application, the termination request is reviewed by Admin before forwarding it to ALK for final decision. ALK then decide to approve or reject the application.

After making decisions for membership, loan and membership termination application, the system will update the applicant's status, notify the applicants and the status is passed to admin for further report.

2.2 System Features

The Koperasi KADA Automated System is a software solution designed to enhance operation efficiency. Access to the system is possible on standard desktop and mobile devices, operating on widely used browsers and operating systems. It puts at the disposal of the staff to manage the membership, loan and termination application more effectively. In changing from manual procedures to an automated system, the new system guarantees in increasing the efficiency of operation, security of data and more user-friendly in usage for all relevant parties. The system features a real-time database, integrated for easy tracking and management applications, with ease of access both for applicants and KADA's staff. The system features are as follows ;

1. Membership Management

Description :

- The system supports membership registration, review membership, membership approval and record membership data.

Requirements :

- Allow secure submission and review of membership applications
- Allow ALK to approve or reject membership applications
- Allow applicants to monitor the status of their application in real-time
- Record the data

2. Loan Management

Description :

- The system supports loan application, loan review and loan approval as well as record loan data.

Requirements :

- Allow members to submit loan application, admin to review and ALK to approve loan application
- Allow applicants to monitor the status of their application in real-time
- Record the data

3. Terminate Membership Management

Description :

- The system supports a termination membership application, review and

approval as well as record data.

Requirements :

- Allow members to submit terminate membership applications, admin to review and ALK to approve the application.
- Allow applicants to monitor the status of their application in real-time
- Record the data

4. Report Generator

Description :

- Generate detailed report for members

Requirements :

- Allow administrators to generate detailed financial statements and reports for Koperasi KADA members focusing on savings balance and loan balance

5. Notification

Description :

- Automated email notification related to loan and membership status

Requirements :

- Sends automated email notification to applicants regarding membership and loan status of their applications
- Sends generated financial statements through email notification

6. Financial Transaction

Description :

- A system for recording, updating, and reconciling financial transactions

Requirements :

- Allow administrators to records all saving transactions in the koperasi
- Allow administrators to keep track of all loan repayments

7. Status Modification

Description

- A function to change membership and loan status

Requirements :

- Allow ALK to change membership status from Diterima to Ditolak and vice versa
- Allow ALK to change loan status from Diterima to Ditolak and vice versa

2.2.1 Use Case

Figure 2.1 below is a use case that illustrates the structure of the Koperasi KADA Automated System. This use case highlights key modules such as membership application module, loan application module, review module, approval module and statement report production module. Each module covers a particular use case such as LogIn to Apply Membership (UC01) in the membership application module, Apply for Loan (UC04) and Calculate Loan and Dividend (UC05) in loan application module, Review Applicant's Form (UC02) in review module, Approve Membership (UC03) and Approve Loan Application (UC06) in approval module, Prepare Statement and report (UC07) and Notify via Email (UC08) in statement report generation module as well as Manage Transactions (UC09) in financial tracking module. The detailed description of each module and functions is tabulated in Table 2.1.

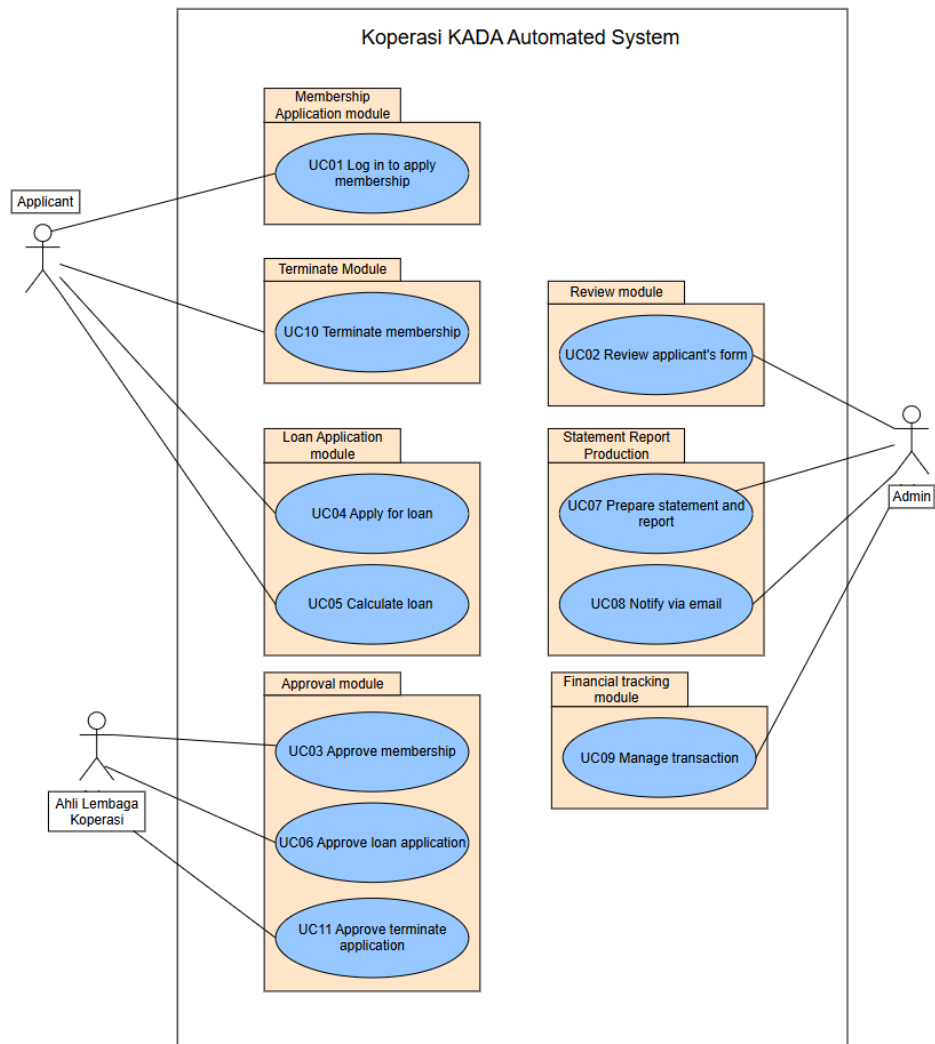


Figure 2.1: Use Case Diagram for Koperasi KADA's Automated System

Table 2.1: Description of Module and Functions for Koperasi KADA's Automated System

Use Case	Function	Description
UC01	Login to apply membership	This use case enables applicants to login, complete and submit a membership form within the system.
UC02	Review applicant's form	This use case ensures the admin to thoroughly check and validate each applicant's form before final submission.
UC03	Approve Membership	This use case enables Ahli Lembaga Koperasi (ALK) to approve or reject membership applications submitted

		by applicants.
UC04	Apply Loan	This use case enables members to submit loan applications online.
UC05	Calculate Loan and Dividend	This use case automates the calculation of monthly repayment and applicable dividends based on applicant's input.
UC06	Approve Loan Application	This use case enables the Ahli Lembaga Koperasi (ALK) staff to review and make a decision on loan applications submitted by Koperasi members.
UC07	Prepare Statement and Report	This use case allows the system to generate a detailed statement report for loan applicants based on their approved loan applications.
UC08	Notify via Email	This use case enables the system to send notifications to applicants regarding their loan statement and other important updates.
UC09	Manage Transaction	This use case enables the system to handle and record all savings and loan repayment transactions whether on a monthly or irregular basis efficiently.
UC10	Terminate Membership	This use case enables the applicant to request membership termination.
UC11	Approve Terminate Application	This use case enables the ALK to approve or reject membership termination applications for members that wish to end their membership.

2.2.2 Domain Model: Class Diagram

Figure 2.2 illustrates a domain model representing the key entities and relationships in a membership and loan management system. The model includes users, administrators, and ALK staff, who manage membership applications, approvals, and modifications. Applicants provide personal, family, and bank information when applying for membership, loans and membership terminations. The system tracks loan details,

```

classDiagram
    class User {
        -email: String
        -password: String
        -firstName: String
        -lastName: String
        -created_at: String
    }
    class UserType {
        -userType: int
        -userStatus: String
    }
    class Managing {
        -status: String
        -notificationLatestDate: String
    }
    class Administrator {
        -staffNo: String
    }
    class ALK {
        -staffNo: String
    }
    class Applicant {
        -staffNo: String
        -applicantName: String
        -applicantIC: String
        -applicantPF: String
        -applicantAddress: String
        -applicantStreet: String
        -applicantPostcode: String
        -applicantCity: String
        -applicantState: String
        -applicantGender: String
        -applicantReligion: String
        -applicantRace: String
        -applicantPosition: String
        -officeAddress: String
        -officeStreet: String
        -officePostcode: String
        -officeCity: String
        -applicantPhoneNumber: int
        -applicantDOB: int
        -applicantAge: int
        -maritalStatus: String
        -applicantGrade: String
        -applicantFax: int
        -applicantPhoneHome: int
        -applicantSalary: double
        -salaryStatement: String
    }
    class Membership {
        -membershipStatus: String
        -membershipApplyDate: String
        -membershipApproveDate: String
        -membershipReviewDate: String
    }
    class MembershipEnd {
        -formID: int
        -applyDate: String
        -endDate: String
        -reason: String
        -reviewDate: String
        -approveDate: String
    }
    class Status {
        -status: int
        -statusDesc: String
    }
    class historyModifiedMember {
        -historyMemID: int
        -newMemStatus: String
        -memChangeDate: String
        -memModifiedBy: String
    }
    class Loan {
        -loanID: String
        -loanAmount: double
        -loanDuration: int
        -loanType: int
        -monthlyPayment: double
        -bankName: String
        -accountBankNumber: String
        -applicantMonthlyNetSalary: double
        -applicantMonthlyBasicSalary: double
        -verificationLetter: String
        -loanApproveDate: String
        -otherLoan: String
        -loanReviewDate: String
        -sendStatus: int
    }
    class LoanType {
        -loanType: int
        -loanName: String
    }
    class AdminApplicationFeedback {
        -feedbackID: int
        -email: String
        -reason: String
        -dateGiven: String
    }
    class AlkApplicationFeedback {
        -feedbackID: int
        -email: String
        -reason: String
        -dateGiven: String
    }
    class historyModifiedLoan {
        -historyLoanID: int
        -newLoanStatus: String
        -loanChangeDate: String
        -loanModifiedBy: String
    }
    class Saving {
        -savingAmount: double
        -savingDate: String
        -savingDesc: String
        -savingType: String
        -savingReceipt: String
        -savingConfirm: String
    }
    class FamilyInfo {
        -familyIC: String
        -count: int
        -relationship: String
        -familyName: String
    }
    class BankInfo {
        -accountBankNumber: String
        -bankAccountName: String
        -bankStatement: String
    }
    class savingType {
        -freeMasuk: double
        -modalSyer: double
        -modalYuran: double
        -wangDepositiAnggota: double
        -sumbanganTabung: double
        -simpananTetap: double
        -lainLain: double
    }
    class Guarantor {
        -guarantorIC: String
        -guarantorName: String
        -guarantorID: String
        -guarantorPF: String
        -guarantorPhoneNumber: String
    }
    class LoanRepayment {
        -repaymentID: int
        -repaymentAmount: double
        -repaymentDate: String
        -repaymentDesc: String
        -repaymentType: String
        -repaymentReceipt: String
        -repaymentConfirm: String
    }

    User "1" *-- "1..*" User
    Managing "1..1" --> "1..1" User : Get
    Managing "0..*" --> "1..1" Administrator : Send
    Administrator "1..1" --> "1..1" Membership : Review
    Administrator "1..1" --> "1..1" ALK : Review
    ALK "1..1" --> "1..1" Applicant : Approve
    ALK "1..1" --> "1..1" MembershipEnd : Approve
    Applicant "1..1" --> "1..1" Membership : Apply
    Applicant "1..1" --> "1..1" Loan : Apply
    Applicant "1..1" *-- "1..1" Saving
    Applicant "1..1" *-- "1..1" FamilyInfo
    Applicant "1..1" *-- "1..1" BankInfo
    Applicant "1..1" *-- "1..1" savingType
    Applicant "1..1" *-- "1..1" Guarantor
    Membership "1..1" *-- "1..1" Status
    Membership "1..1" *-- "1..1" historyModifiedMember
    Membership "1..1" --> "1..1" Loan : Apply
    Status "1..1" --> "1..1" Loan : Modify
    historyModifiedMember "0..*" --> "1..1" Loan : Modify
    Loan "1..1" *-- "1..1" LoanRepayment
    Loan "1..1" --> "1..1" LoanType
    AdminApplicationFeedback "0..1" --> "1..1" Loan : Get
    AlkApplicationFeedback "0..1" --> "1..1" Loan : Get
    historyModifiedLoan "0..1" --> "1..1" Loan : Get
    Loan "1..1" --> "1..1" LoanRepayment : Modify
  
```

2.2.3 State Diagram

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applicant completes the loan application and becomes the 'Loan Under Review' state where the ALK needs to approve the application. If the loan is rejected the applicant can reapply the application. If the loan has been approved, the loan status will change to 'Loan Active'. When the loan is fully paid, the process goes to 'Loan Completed', which closes the loan.

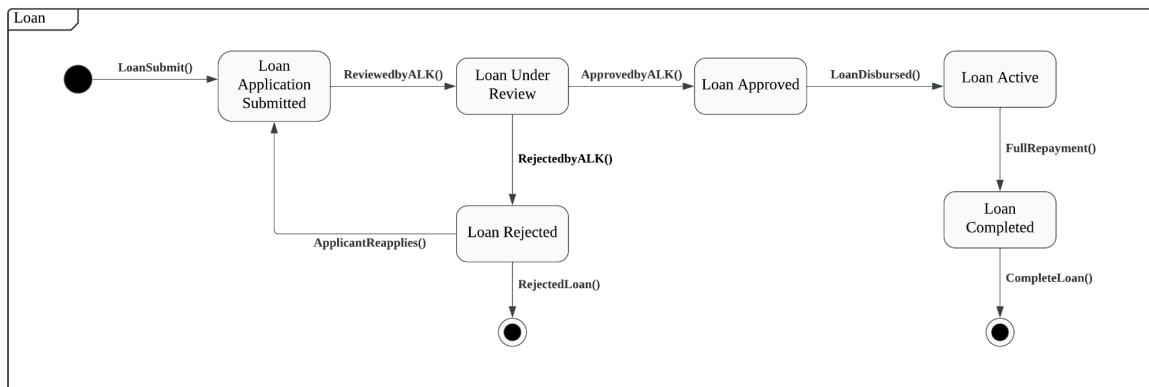


Figure 2.3: State Machine Diagram of Loan for Koperasi KADA's Automated System

Figure 2.4 shows the Membership State Machine Diagram's sequence of an application for membership. The process begins when a user submits the application, and it goes to 'Membership Under Review', the ALK either approves it or denies it. If approved, the membership becomes 'Membership Active'. From here, applicants can allow the membership to lapse and be 'Membership Expired'. The rejected applications enable applicants to reapply and go through the process all over again.

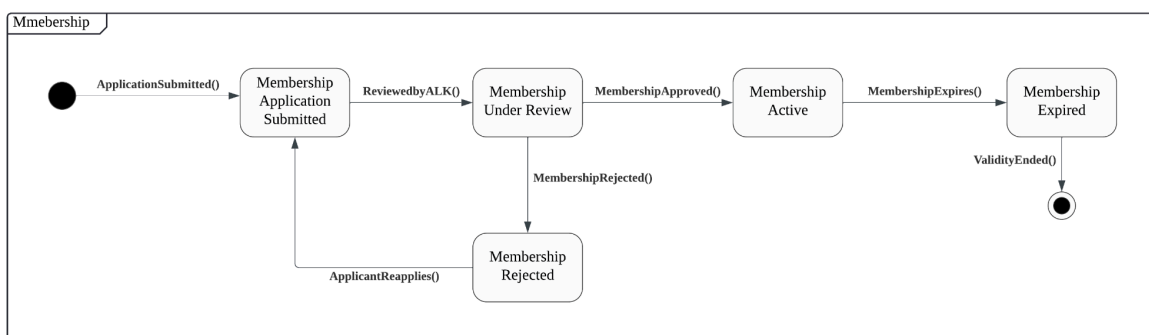


Figure 2.4: State Machine Diagram of Membership for Koperasi KADA's Automated System

Figure 2.5 shows the Report Generation State Machine Diagram. This is the process of creating and distributing the report. It starts with 'Report Initiated' each time the Admin initiates the creation of a report. After being generated, the report can go to the recipient and be in the state of 'Report Sent' or be stored in the system at the state of 'Report Stored'.

Notifications are sent with the report, and status changed to 'Notification Created' and when the recipient reads the notification, it ends at 'Notification Read'

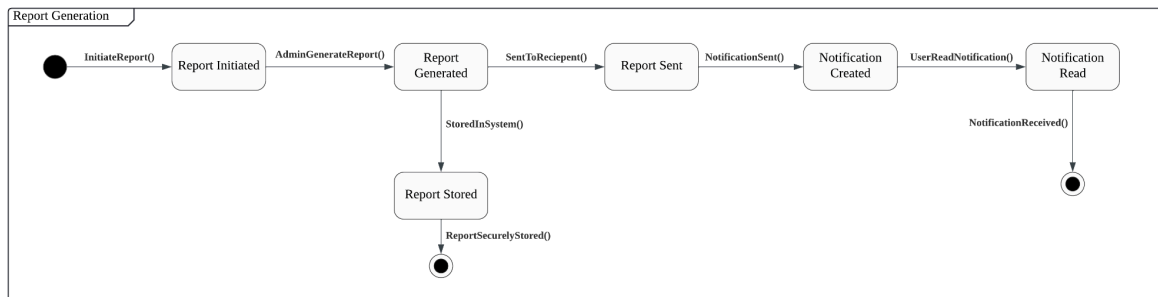


Figure 2.5: State Machine Diagram of Report Generation for Koperasi KADA's Automated System

Figure 2.6 shows the Membership Termination State Machine Diagram. The process begins when a member submits a termination request, entering the 'Membership Termination Application Submitted' state. The request is then reviewed, leading to either 'Membership Termination Approved' or 'Membership Termination Rejected'. If rejected, the applicant can reapply, restart the application process. Once approved, the membership ends and moves to 'Membership Termination Completed', concluding when validity expires.

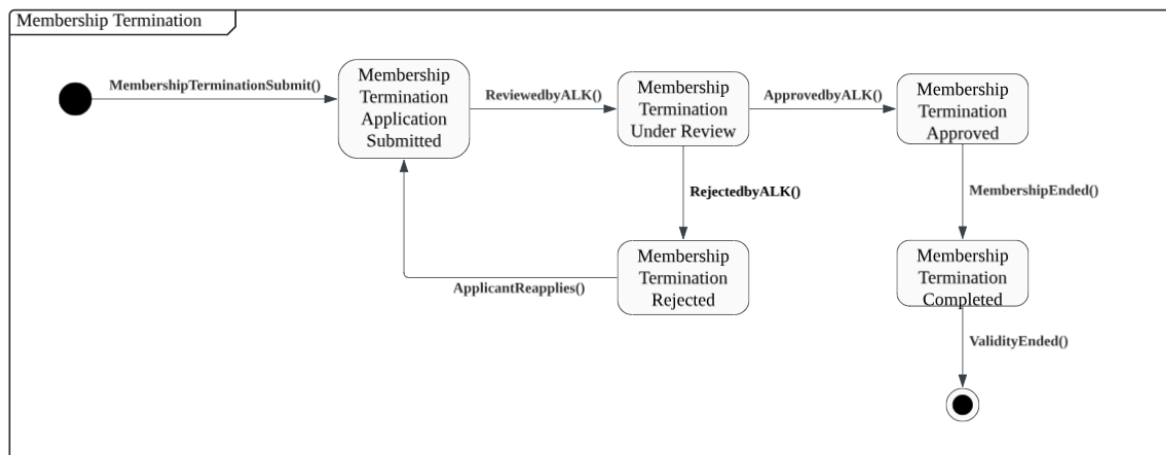


Figure 2.6: State Machine Diagram of Membership Termination for Koperasi KADA's Automated System

UC001: Use Case Login to Apply Membership

Table 2.2 user story description allows applicants to access the membership application system securely. After entering valid login credentials, users are redirected to a dashboard where they can apply for membership by filling out a detailed application form. The system verifies the submitted information and ensures all required fields are completed. If the

information is valid, the application is successfully submitted. If any issues are detected, the system provides feedback to guide users in correcting their errors.

Table 2.2: Use Case Description for Login to Apply Membership

Use Case: Login to Apply Membership
ID: UC001
User Story Description As an applicant, I want to log into the membership application system, So that I can apply for membership.
Flow of events: 1. The Applicant navigates to the login page of the system. 2. The system prompts the Applicant to enter their username and password. 3. The Applicant enters their login details and submits the form. 4. The system verifies the login information. 4.1 If the login details are correct, the system grants access to the Applicant. 4.2 If the login details are incorrect, the system displays an error message. 5. Once logged in, the Applicant can access the membership application form.
Alternative flow n: n.1 : If the applicant forgets their login details, they can click on the “Forgot Password” option. n.2 : The system guides the Applicant through a password recovery process by sending a recovery link to their registered email.
Acceptance Criteria Postcondition : 1. The Applicant is successfully logged into the system and gains access to the membership application form. Precondition : 1. The Applicant must be registered in the system with a valid username and password 2. The system must be accessible for login attempts. Other conditions : 1. The system must ensure that all login details are encrypted for security purposes. 2. The login page must support both desktop and mobile devices.
Exception flow: 1. If the Applicant enters invalid login details three times, the system locks the account temporarily and prompts Applicant to use the “Forgot Password” option. 2. If the system is unavailable due to maintenance, the Applicant can see a maintenance notice and cannot login until the system is recovered.

Figure 2.7 illustrates the applicant's login process. The applicant accesses the login page, enters their details and the system will validate the details. If the details are valid, access to the membership application form is granted. Otherwise, an error message is displayed. The membership form is then accessed and shown to the applicant.

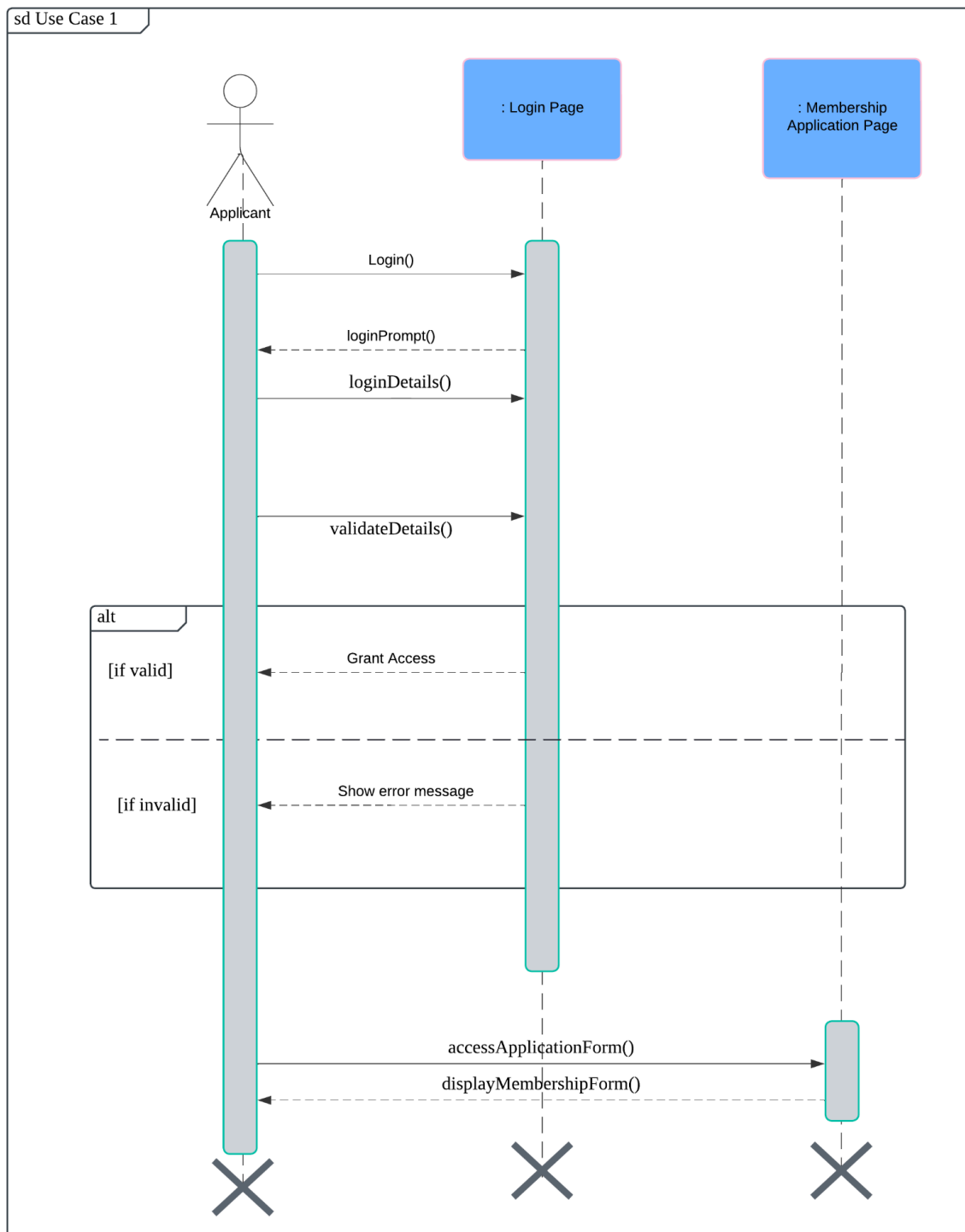


Figure 2.7: Sequence Diagram for Login to Apply Membership

UC002: Use Case Review Applicant's Form

Table 2.3 user story description allows administrators to review multiple membership, loan, or membership termination applications in batches. The process begins with the administrator selecting a batch of applications for review. Based on the review, the admin can approve or reject the application based on the provided information. For rejected applications, the system allows the admin to provide specific reasons for rejection. Approved applications are sent to the next stage of processing.

Table 2.3: Use Case Description for Review Applicant's Form

Use Case: Review Applicant's Form
ID: UC002
User Story Description As an admin, I want to review the membership, loan and membership termination application forms submitted by applicants, So that I can verify applicant details and determine whether the forms are complete or incomplete for further processing.
Flow of events: 1. The Admin logs into the system. 2. The Admin navigates to the section containing submitted membership, loan or membership termination 3. The system displays a list of all pending applications. 4. The Admin selects an application form to review. 5. The system displays the details of the selected application. 6. The Admin reviews the information provided by the Applicant.
Alternative flow <i>n</i>: n.1 : If the Admin encounters an incomplete or invalid application form : 1. The Admin flags the application for follow-up and adds a note specifying the missing or incorrect information. n.2 : If the Admin needs to pause the review and continue later : 1. The Admin saves the progress of the review without finalizing the status. n.3 : If the Admin accidentally selects the wrong application : 1. The Admin uses the "Back" button to return to the list of applications.
Acceptance Criteria Postcondition :

1. The admin successfully reviews the application form and determines the validity of applicant's details.
Precondition : 1. The Admin must have valid login details to access the system. 2. Applicants must have submitted their membership applications forms for review. Other conditions : 1. The system must allow the Admin to view all details of the selected application form. 2. The system must ensure that only pending applications are displayed for review.
Exception flow: 1. If the system is down, the Admin cannot access the application forms and is shown a maintenance message. 2. If an application form is incomplete or contains errors, the Admin flags it for further action.

Figure 2.8 shows the process of an admin reviewing membership, loan and termination membership applications. The admin logs in, navigates to the application list, and views batch application details. The system loops through each application in the batch, marking it as either “reviewed” or “incomplete”.

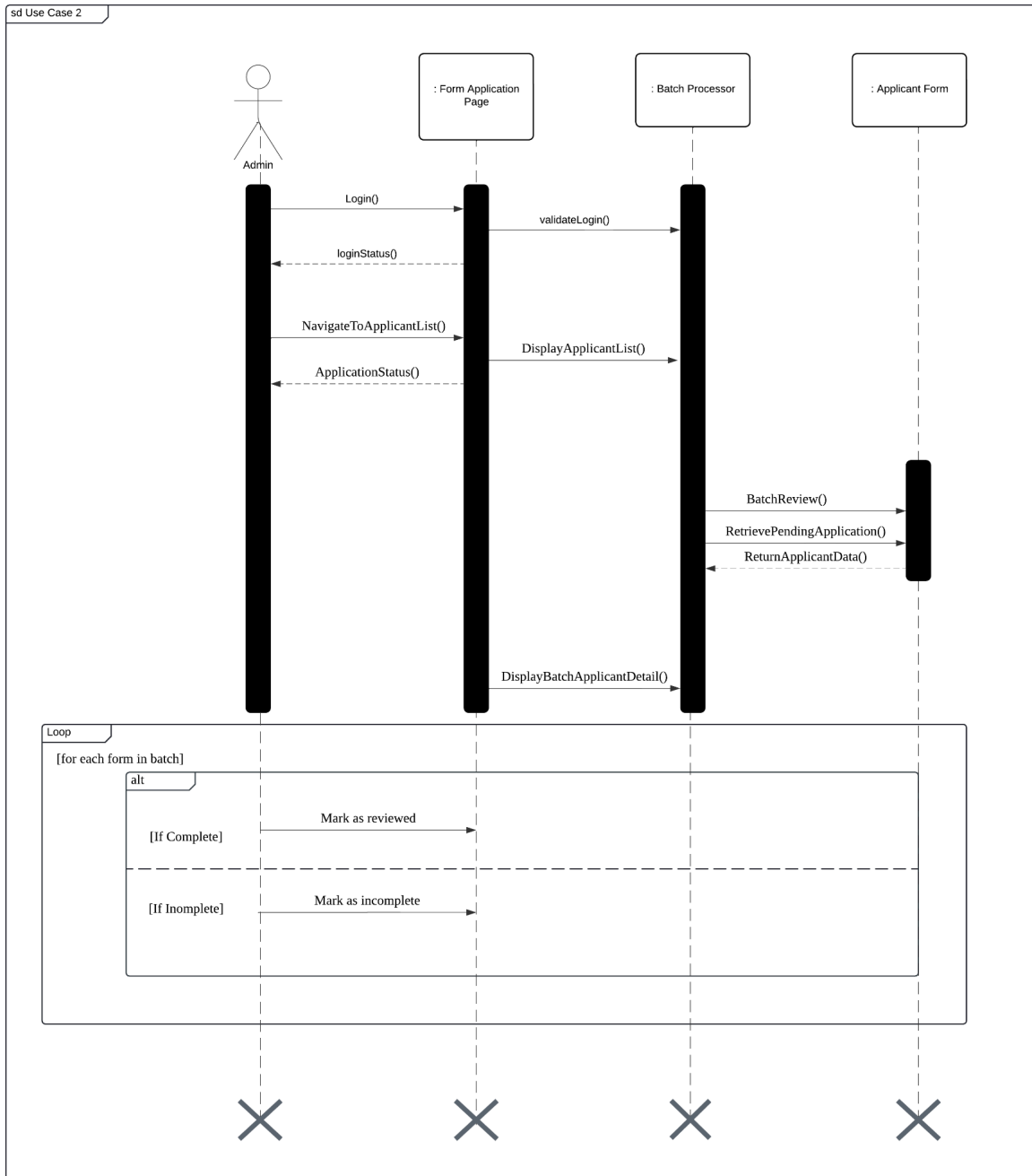


Figure 2.8: Sequence Diagram for Review Applicant's Form

UC003: Use Case Approve Membership

Table 2.4 describes the process of ALK in approving membership applications. ALK accesses the pending applications, assesses them and makes a decision on their approval

or rejection, basing their judgement on the applicants' eligibility. If applications are incomplete, they are rejected. If further discussion is required, they are flagged for clarification. When a decision is reached, the system will update the applicant's status regarding membership and record this. The preconditions are that the admin must view the applications beforehand and ALK would have logged on. System shall ensure applicant documents access is secure.

Table 2.4: Use Case Description for Approve Membership

Use Case: Approve Membership
ID: UC003
User Story Description As an Ahli Lembaga Koperasi (ALK), I want to approve applicants' membership , So that I can ensure that only eligible individuals can become membership and earn membership benefits
Flow of events: 1. ALK accesses the pending list of membership applications 2. ALK chooses a membership applications to evaluate 3. ALK can choose batch processing for approval simultaneously 4. ALK makes decision to approve or reject the membership application 5. ALK chooses next membership application
Alternative flow <i>n</i>: n.1 : If the application status forwarded by admin is "Incomplete", ALK will reject immediately n.2 : If ALK requires additional discussion, the application is flagged for further clarification in meeting
Acceptance Criteria Postcondition : 1. The applicant's membership status is updated. 2. The system records the applicant's membership status. Precondition : 1. Admin has reviewed the membership applications. 2. The ALK must be logged into the system. Other conditions : 1. The system must allow ALK to view all documents uploaded by applicants
Exception flow: 1. If the system is unavailable, both admin and ALK cannot process applications and error message will be displayed

Figure 2.9 shows a sequence diagram for use case approve membership. ALK logs into the system, which either grants access or reuters an error. ALK then selects any membership application from the pending list, evaluates and decides whether to approve or not. Finally, the system updates the loan status

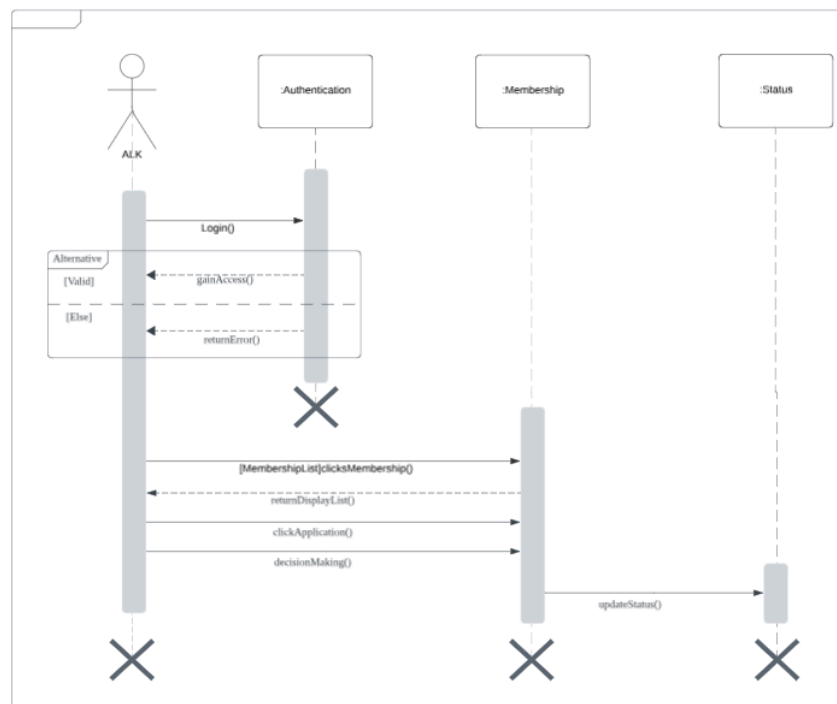


Figure 2.9: Sequence Diagram for Approve Membership

UC004: Use Case Apply Loan

Table 2.5 user story description emphasises on enabling members of Koperasi KADA to submit loan applications seamlessly through an online platform. The process starts with the applicant filling out the compulsory details in the loan application form. The system validates the application to ensure all required information is correct and complete once it is submitted. If the validation is successful, the application is stored in the loan database for further processing. Applicants will receive a confirmation message after the submission and can track their application status in real-time. This user story description plans for the data accuracy while aims to provide an outlined and user-friendly exposure for loan applications.

Table 2.5: Use Case Description for Apply Loan

Use Case: Apply Loan
ID: UC004
User Story Description As an member of Koperasi KADA, I want to apply for loan through the Koperasi KADA Automated System, So that I can request financial support for my needs.
Flow of events: 1. The member logs into the system by accessing the login page and entering valid login credentials. 2. The member selects the “Apply Loan” option after logged in. 3. The system displays the loan application form. 4. The member completes the loan application form. 1. The member submits the loan application form to the system.
Alternative flow <i>n</i>: n.1 : If any required fields are empty or invalid, the system will ask the member to re-enter the correct or valid information before submit.
Acceptance Criteria Postcondition : 1. The loan application is successfully submitted and recorded in the system. 2. The member receives a confirmation notification. Precondition : 1. The member must be logged into the system. 2. The member must have an approved membership status. Other conditions : 1. The member must meet loan requirements as defined by Koperasi KADA.
Exception flow: 1. If the system detects a problem with the member’s account, an error message is displayed and the members cannot proceed with the application.

Figure 2.10 is the sequence diagram for the application loan process, representing the interaction between the applicant and the system for applying for the loan. It starts when the user logs in and accesses the loan application page. The user fills out the loan application with compulsory details, then the system validates it. If the details are correct, the application is submitted and saved into the system’s database. The system notifies the user after accepting the successful submission. This diagram illustrates the important steps involved, including user interactions, form validation and database operations.

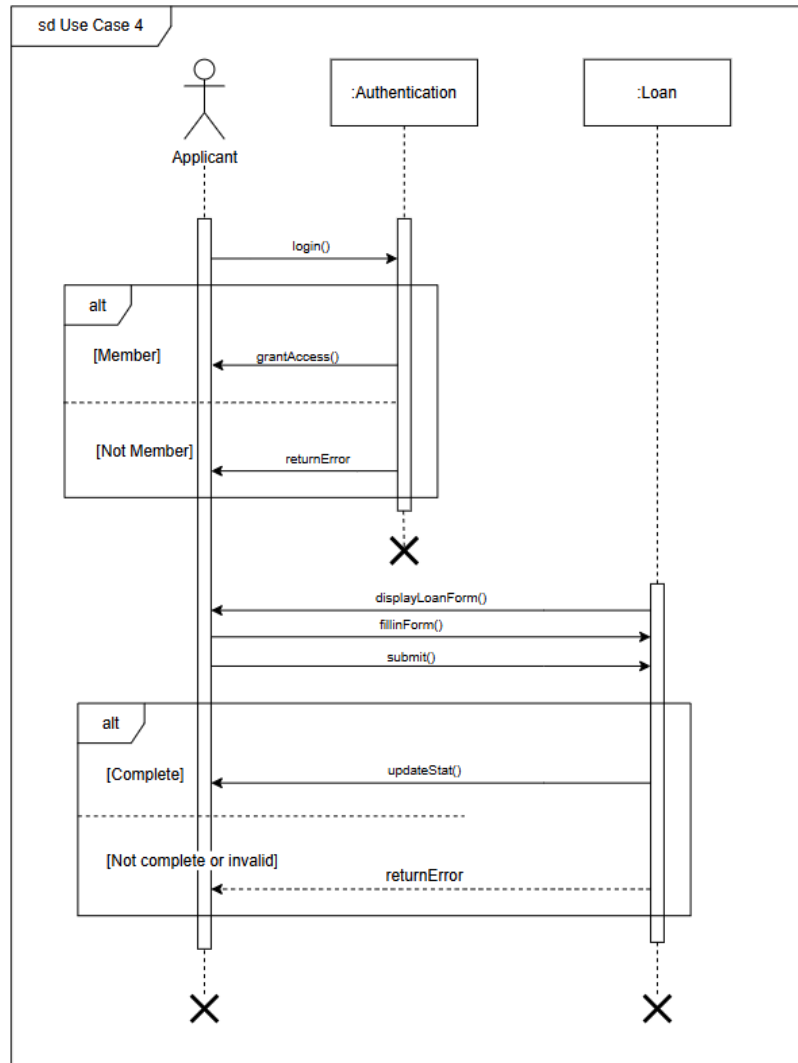


Figure 2.10: Sequence Diagram for Apply Loan

UC005: Use Case Calculate Loan

Table 2.6 focuses on the members of Koperasi KADA who want to calculate their loan repayment and dividend returns to know about their financial commitments and benefits. The table outlines the process that includes the member logging into the system, selects the “Loan Calculator” button features, enters the loan details like amount, duration and interest rate. Then, click the “Calculate” button and the system validates the inputs. If valid, the system displays the repayment results. If invalid, it shows an error message.

Table 2.6: Use Case Description for Calculate Loan and Dividend

Use Case: Calculate Loan and Dividend
ID: UC005
User Story Description As an member of Koperasi KADA, I want to calculate my loan repayment and dividend returns, So that I know about my financial commitments and also benefits.
Flow of events: 1. The member logs into the system by accessing the login page and entering valid login credentials. 2. The member selects the "Loan Calculator" option after logged in. 3. The system displays the loan calculator page. 4. The member enters the loan desired amount, duration. 5. The member chooses the interest rate from the list of chosen loan types. 6. The member clicks on the "Calculate" button.
Alternative flow <i>n</i>: n.1 : If the member enters invalid amount, an error message will be displayed.
Acceptance Criteria Postcondition : 1. The system calculates and displays the amount of loan to the member. Precondition : 1. The member must be logged into the system. Other conditions : 1. The member must meet loan requirements as defined by Koperasi KADA.
Exception flow: 1. If the system is unable to process calculations, an error message will be displayed.

Figure 2.11 is a sequence diagram for calculating loan and dividend highlights the interaction between members and the system. The process begins with the member logging into the system by entering their details. The system then verifies the details. If valid, the system grants access to the members. If invalid, the system returns an error message. The member selects the "Loan Calculator" button, and the system rate then submits the details. The Loan Calculator calculates and generates the loan repayment by processing these inputs. If the inputs are valid, the system displays the results. If invalid, it returns an error message that requires the member to reenter the inputs. The interaction makes sure the member has correct financial calculations based on their provided details.

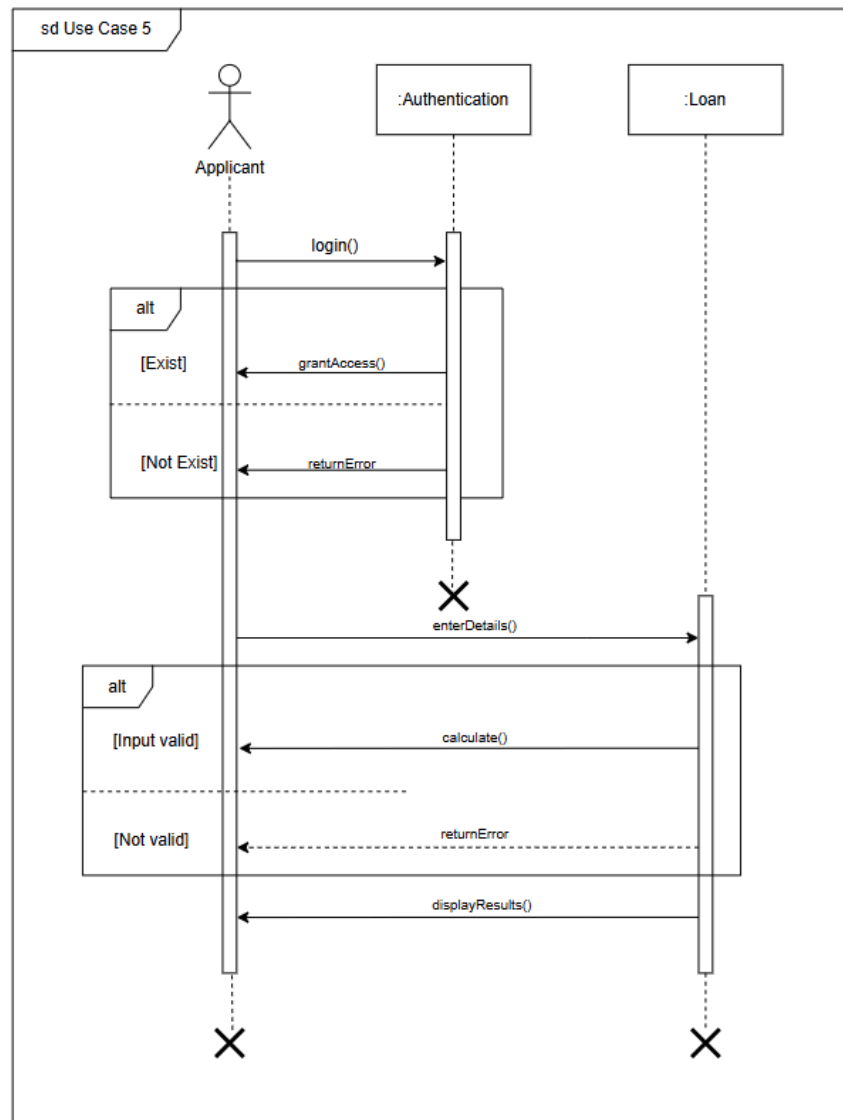


Figure 2.11: Sequence Diagram for Calculate Loan and Dividend

UC006: Use Case Approve Loan Application

Table 2.7 shows a use case description of the procedure of loan application approval by ALK. The system is viewed for pending applications by ALK, evaluates application based on the company's policies, after which a decision to either approve, reject or use the compare feature before finalizing. Then, ALK can proceed to the next application or log out. Alternative flows include rejecting incomplete applications reviewed by admin or postponing decisions. Postconditions ensure the loan status is updated, recorded and applicants

notified. Preconditions required admin's prior review and ALK's login. Exception flows address system downtime or missing data, returning applications to admin for clarification.

Table 2.7: Use Case Description for Approve Loan Application

Use Case ID: UC006
User Story Description: As an Ahli Lembaga Koperasi I want to approve loan application So that i can let only the eligible applicants to receive loan in line with company's policies
Flow of Events: 1. The ALK staff member accesses the list of pending loan applications. 2. The ALK selects a specific loan application to review the applicant's details 3. The ALK staff member evaluates the application based on established criteria. 4. The ALK makes a decision to either directly approve or reject or use the compare button to compare the applicant's form with loan terms before approving or rejecting the loan form. 5. The ALK chooses the next application or log out of the system.
Alternative Flow n: n.1 : If the application is marked "Incomplete" by Admin, ALK will reject immediately n.2 : If ALK need more time to review particular application, ALK will press press back button and do decision later
Acceptance Criteria: Postcondition: 1. Loan application status is updated 2. The system record the loan application status 3. Applicants are notified of the status Precondition: 1. The Admin has reviewed the loan applications 2. ALK must have logged into the system Other condition: 1. All applicants' data must be encrypted and only accessible to authorized personnel
Exception Flow: 1. If the system is down, ALK cannot evaluate the applications, causing delays 2. If ALK cannot make a decision due to missing data, the application is sent back to the Admin for further discussion

Figure 2.12 shows a sequence diagram for use case approve loan applications. ALK logs into the system, which either grants access or reuters an error. ALK then selects any loan application from the pending list, evaluates and decides whether to approve or not. If necessary, ALK uses the compare button to compare against loan terms before finalizing the decision. Finally, the system updates the loan status.

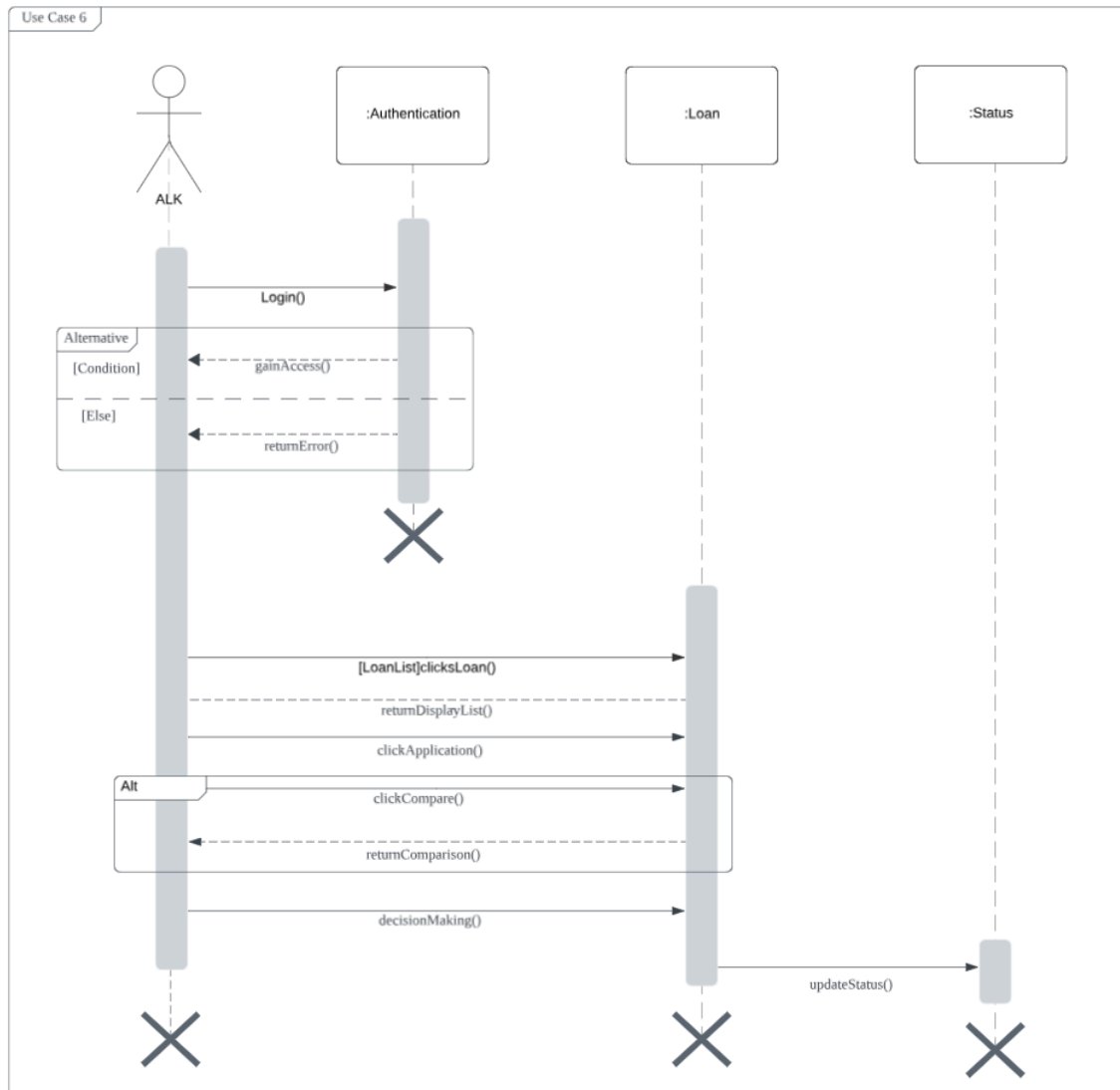


Figure 2.12: Sequence Diagram for Approve Loan Application

UC007: Use Case Prepare Statement and Report

Table 2.8 provides a detailed description of the use case for the Prepare Statement and Report feature, which describes how an admin generates statements and reports to provide applicants with details about their application status, loan history, and membership information. The admin can choose between a financial statement or a corporation report.

The system retrieves relevant data from the database and processes the report based on the selected member. The generated report is then saved for future reference. If the database is unavailable or a system error occurs, the admin will be notified and given options to retry or troubleshoot the issue.

Table 2.8: Use Case Description for Prepare Statement and Report

Use Case: Prepare Statement and Report
ID: UC007
User Story Description As an admin, I want prepare statements and reports for applicants, So that I can provide them with detailed information about their application status, loan history, and membership details.
Flow of events: 1. Select whether the report is for financial statement or corporation report 2. If the report is for generated statement: 2.1. Admin retrieves data for approved member and loan. 2.2. The system fetches the relevant applicant data from the database. 2.3. Admin clicks the "Generate Statement" button to initiate the statement preparation process. 2.4. The system prepares the statement report based on the retrieved data. 2.5. The system displays successful report generation. 3. If the report is for corporation: 3.1. The system fetches the relevant information about membership applications, loan applications and terminate membership applications. 3.2. The system displays the information based on monthly and yearly views.
Acceptance Criteria Postcondition : 1. The statement or report is successfully generated and saved in the system for future reference. Precondition : 1. Admin must be logged into the system with the appropriate permissions. 2. Required data for the statement and report must be accessible within the system. Other conditions : 1. The system must have sufficient access control to ensure only authorized admins can generate reports. 2. The admin must be able to view and select the specific applicant for whom the statement or report is to be generated.
Exception flow:

1. If the database is temporarily unavailable, the system will display a message indicating the issue and allow the admin to retry the process later.
2. If the system fails to generate the statement or report due to system errors, the admin will be notified with a detailed error message and instructed on the next steps.

Figure 2.13 shows the sequence diagram for the process of preparing financial statements and reports by an admin, interacting with the Authentication and Report systems. The admin begins by logging in, where successful authentication grants access to the admin page, while failure triggers an error message. Once authenticated, the admin retrieves member data and generates financial statements for selected members. If statement generation is successful, a confirmation message is displayed; otherwise, an error message appears. Additionally, corporate reports containing membership, loan, and termination statistics are generated, providing monthly and yearly financial summaries.

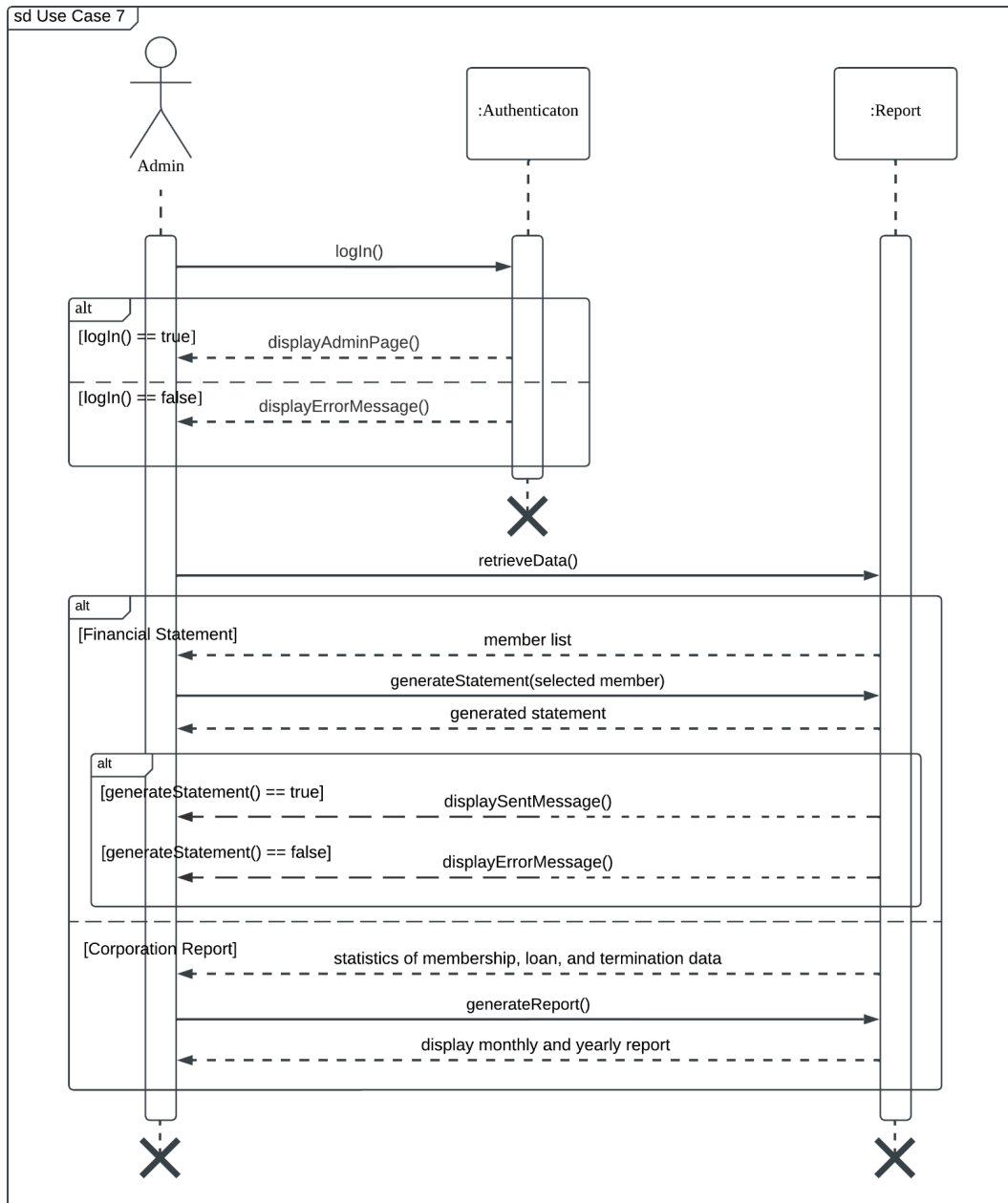


Figure 2.13: Sequence Diagram for Prepare Statement and Report

UC008: Use Case Notify via Email

Table 2.9 describes the Notify via Email use case, where the admin reviews prepared statements and sends notification emails to applicants. The system allows the admin to select whether the email notification is for a generated statement or an updated application status. Once selected, the system generates and sends the appropriate email, logging the

status for tracking purposes. The system also supports batch processing for multiple applicants and handles exceptions such as failed email generation or invalid email addresses by notifying the admin for corrective actions.

Table 2.9: Use Case Description for Notify via Email

Use Case: Notify via Email
ID: UC008
User Story Description As an admin, I want to review prepared statements and send notification emails to applicant, So that I can ensure the applicants are informed about their loan details.
Flow of events: 1. Select whether the email notification is for generated statement or states notification. 2. If the notification is for generated statement: 2.1. The system detects that a loan statement and report have been prepared. 2.2. The Admin reviews the prepared statement and clicks the “Send” button. 2.3. The system generates a notification email with statement details. 2.4. The system sends the email to the selected members. 2.5. The system logs the email status in the admin interface. 3. If the transaction is irregular: 3.1. The Admin tap on “Send” at desired applicants. 3.2. The system generates an email notification based on the updated status. 3.3. The system sends the email to the selected applicants.
Acceptance Criteria Postcondition : 1. Recipients are notified via email, with access to the statement or report as needed. Precondition : 1. The system must have a list of recipients with valid email addresses. 2. The statement or report has been prepared in the system. Other conditions : 1. The system must allow the Admin to view all details of the selected application form before sending the notification email.
Alternative Flow 1: 1. If the Admin wants to batch processing for application status or generated statement. 2. Admin selected and processes multiple applicants at once. 3. The system should then send email notifications to each applicant.
Exception flow: 1. If the system fails to generate a notification email, it will notify the admin of the failure and allow the admin to retry sending the email.

- | |
|--|
| <ol style="list-style-type: none">2. If the email address is invalid, the system will prompt the admin to correct the address before retrying the process. |
|--|

Figure 2.14 illustrates the sequence diagram for the use case Notify via Email, regarding prepared statements and status updates. The Admin initiates the notification process by selecting a notification type, triggering interactions between the Notification and Report components. If a Generated Statement Notification is selected, the system retrieves the prepared statement for the selected member and sends it via email, followed by updating the notification log. Similarly, for Status Notification, an email is sent to the selected applicant, and the send status is updated. The process also supports batch processing, where multiple members or applicants are notified simultaneously. The final step verifies whether the email was successfully sent, displaying either a success or error message accordingly.

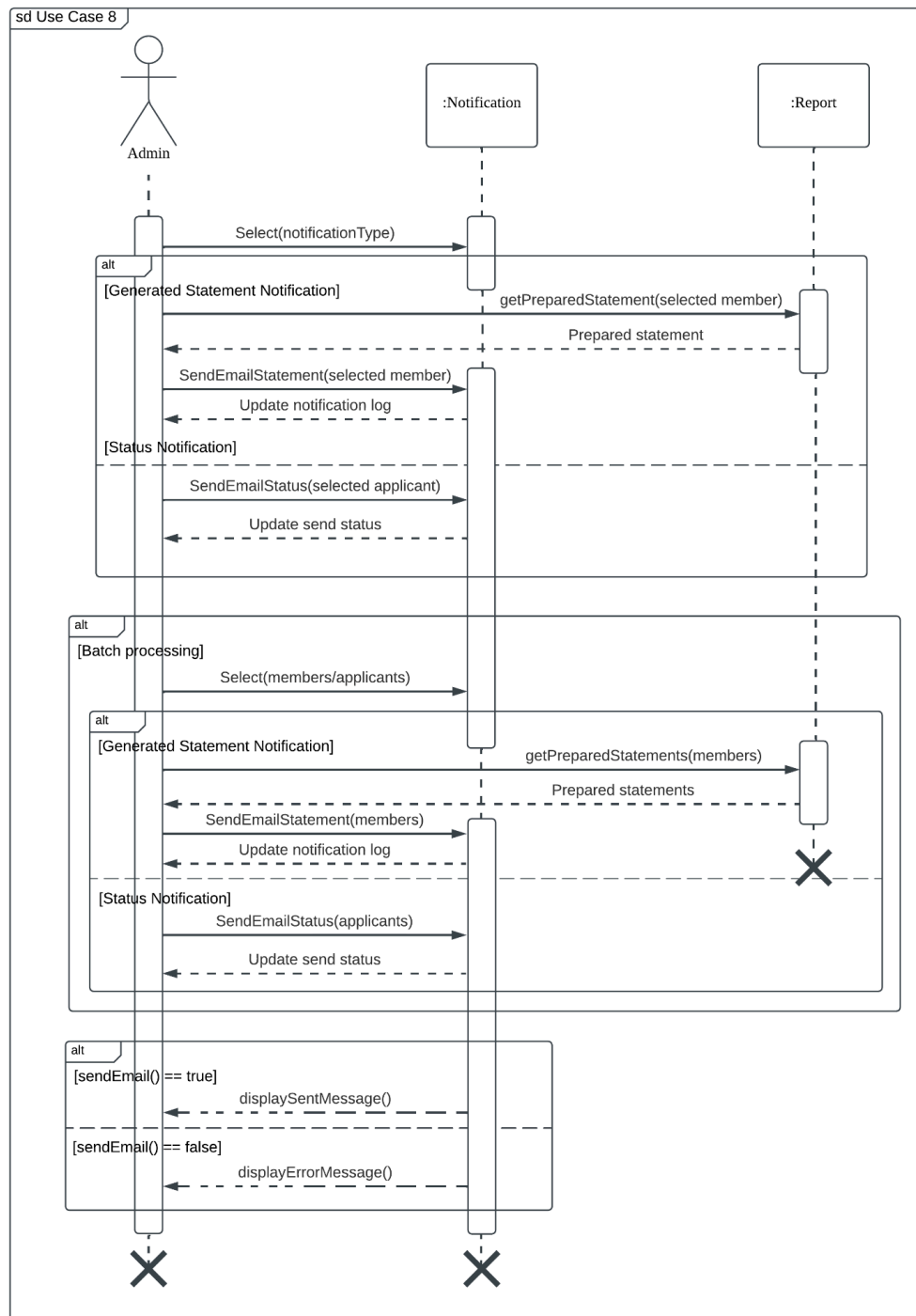


Figure 2.14: Sequence Diagram for Notify via Email

UC009: Use Case Manage Transactions

Table 2.10 describes the Manage Transaction use case. This use case explains how an admin handles and records savings and loan repayment transactions to keep financial

records correct and updated. The admin can choose between monthly or irregular transactions. For monthly transactions, the admin selects a member and processes the update. For irregular transactions, the admin fills out a form and submits it. The system also allows batch processing for multiple monthly transactions. If there are missing details, incorrect member information, or system errors, the system will show a warning and prevent the transaction until fixed.

Table 2.10: Use Case Description for Manage Transactions

Use Case: Manage Transactions
ID: UC009
User Story Description As an admin, I want to manage and record all savings and loan repayment transactions, whether on a monthly or irregular basis, So that I can ensure the financial records are accurate and up to date.
Flow of events: 1. Select whether the transaction is monthly or irregular. 2. If the transaction is monthly: 2.1. Choose the member involved in the transaction and click "Keluarkan". 2.2. The system updates financial transactions. 3. If the transaction is irregular: 3.1. Select Savings or Loan Repayment. 3.2. Fill in the transaction form. 3.3. Submit the form for processing.
Acceptance Criteria Postcondition : 1. The transaction is successfully recorded in the system. Precondition : 1. The admin is logged into the system. 2. The membership application status is approved in the system. Other conditions : 1. Only authorized administrators can process transactions.
Alternative Flow 1: 1. If the Admin wants to batch processing for monthly transactions 2. Admin selected and processes multiple monthly transactions at once. 3. The transactions are recorded and updated.
Exception flow: 1. If required fields are missing in the transaction form, the system prompts an error. 2. If the selected member does not exist, the system prevents transaction processing. 3. If there is a system error, the transaction is logged for review.

Figure 2.15 illustrates the process of managing financial transactions by an admin, interacting with the Transaction and Financial Record systems. The admin selects a transaction type, which can be a Monthly Transaction, an Irregular Transaction, or Batch Processing Monthly. For monthly transactions, the system deducts the salary of a selected member, while irregular transactions involve selecting a member and specifying savings or loan repayment. In batch processing, multiple members are selected for financial updates. Each transaction update is confirmed, and the system displays either a success or error message based on the update status.

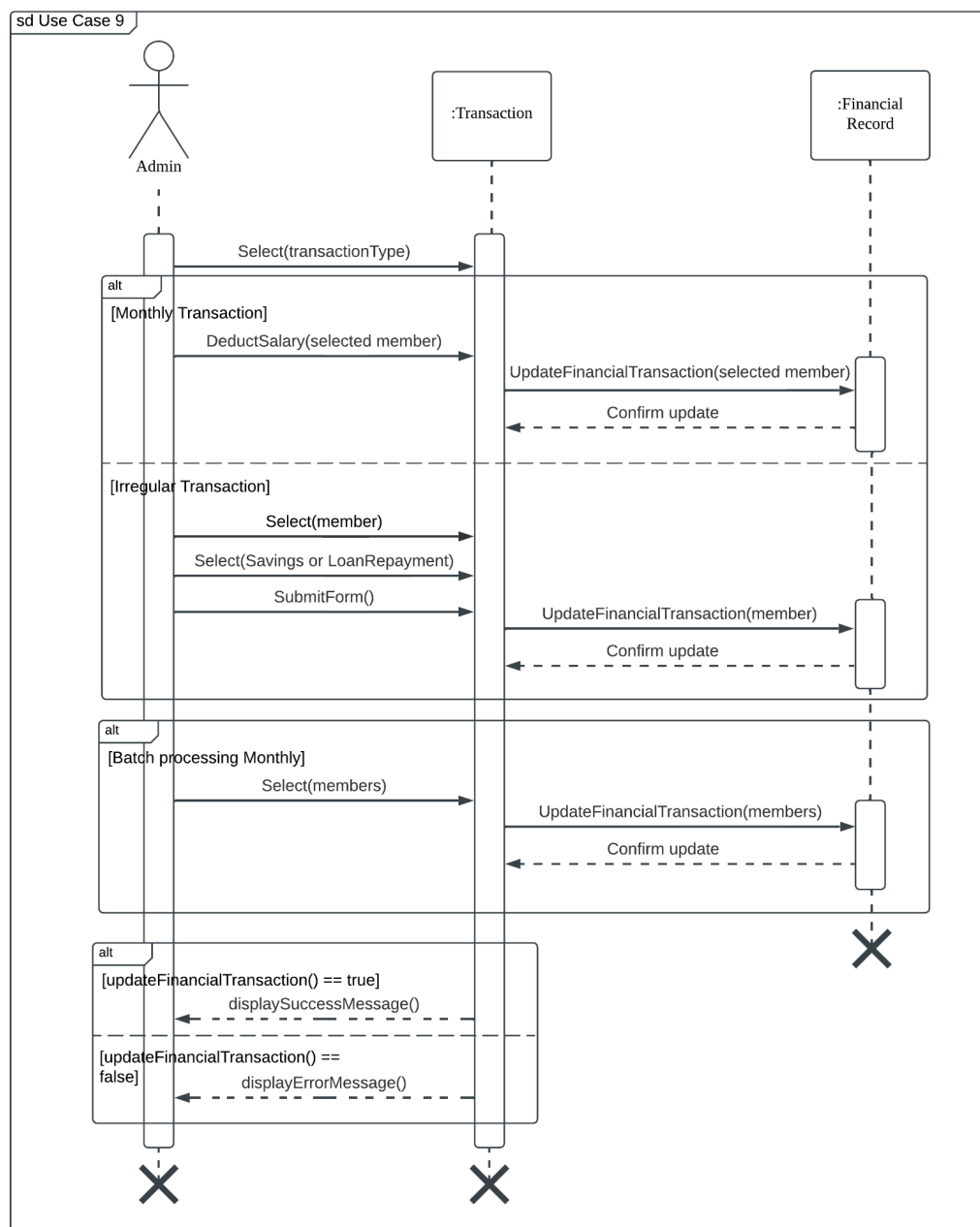


Figure 2.15: Sequence Diagram for Manage Transactions

UC010: Use Case Terminate Membership

Table 2.11 is a use case description that enables an Applicant to request termination of the Koperasi KADA membership. The Applicant submits the required details through the system, which validates the request and then forwards it to the Admin for review. Once they submit the form, the status of the application can be displayed on the status page.

Table 2.11: Use Case Description for Terminate Membership

Use Case ID: UC010
User Story Description: As an Applicant I want to terminate my membership So that i can discontinue my association with the cooperative
Flow of Events: 1. The Applicant logs into the system. 2. The Applicant navigates to the membership termination session. 3. The Applicant submits a termination request by providing the required details. 4. The system validates the termination request for completeness. 5. The system notifies the Applicant of the decision via email or system notification.
Alternative Flow n: n.1 : If the application is marked “Incomplete” by Admin, the system notifies the Applicant to resubmit with all required details.
Acceptance Criteria: Postcondition: 1. The membership termination request is processed. 2. The Applicant is notified of the outcome. Precondition: 1. The Applicant has an active membership. 2. The Applicant is logged into the system. Other condition: 1. All applicants' data must be encrypted and only accessible to authorized personnel.
Exception Flow: 1. If the system is down, the Applicant cannot submit the termination request, causing delays.

Figure 2.16 is a sequence diagram for Terminate Membership use case that illustrates the interaction between the Applicant, Authentication system and Terminate Membership module. The Applicant logs in and then the system verifies their membership

status. If access is granted, the termination form is displayed. The Applicant fills in and submits the form. The system will validate the request, if complete, it updates the status of application. Otherwise, it returns an error to the Applicant.

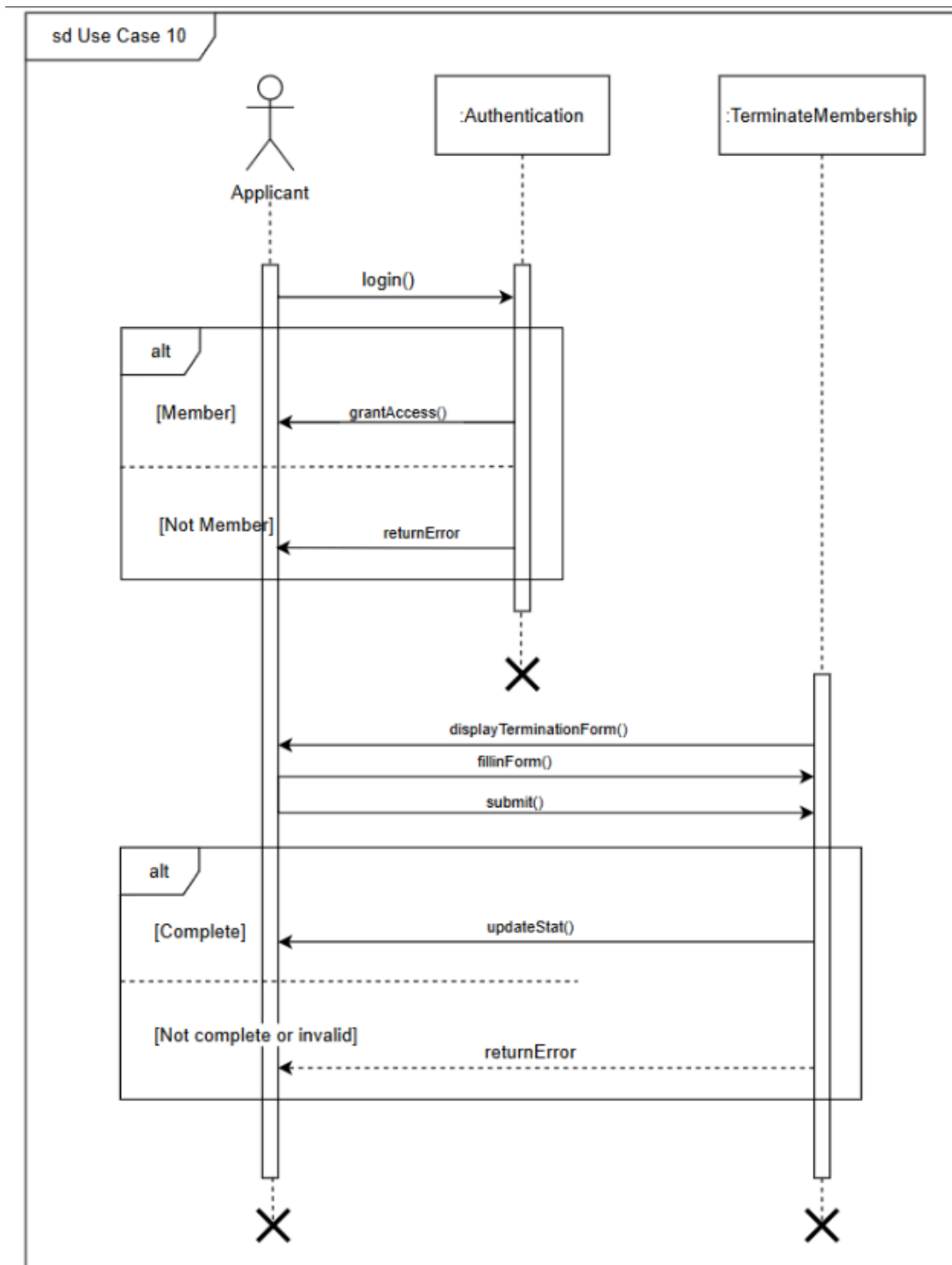


Figure 2.16: Sequence Diagram for Terminate Membership

UC011: Use Case Approve Terminate Application

Table 2.12 is a use case description that enables an ALK to approve termination membership application. The application must be reviewed by Admin beforehand. Then, ALK will make decisions based on circumstances to approve or reject applications. If the application is rejected, then the membership continues as usual otherwise the membership ends immediately.

Table 2.12: Use Case Description for Approve Terminate Application

Use Case ID: UC011
User Story Description: As an Alk I want to approve terminate application So that i can discontinue member's membership
Flow of Events: 1. The ALK logs into the system. 2. The ALK navigates to the terminate application page. 3. The ALK performs decision-making to approve or reject application. 4. The system notifies the Applicant of the decision via email or system notification.
Alternative Flow n: n.1 : If the application is marked "Incomplete" by Admin, the application will not be forwarded to ALK
Acceptance Criteria: Postcondition: 1. Status is updated 2. The Applicant is notified of the outcome. Precondition: 1. Application reviewed by Admin Other condition: 1. All applicants' data must be encrypted and only accessible to authorized personnel.

Figure 2.17 is a sequence diagram for the Approve Terminate Application use case that illustrates the interaction between the ALK and Terminate Membership module. The ALK will log in and then the system will display the termination application reviewed by the admin. The ALK then performs decision-making whether to approve or reject the application. Then the latest status will be updated to keep the applicant informed of real-time status.

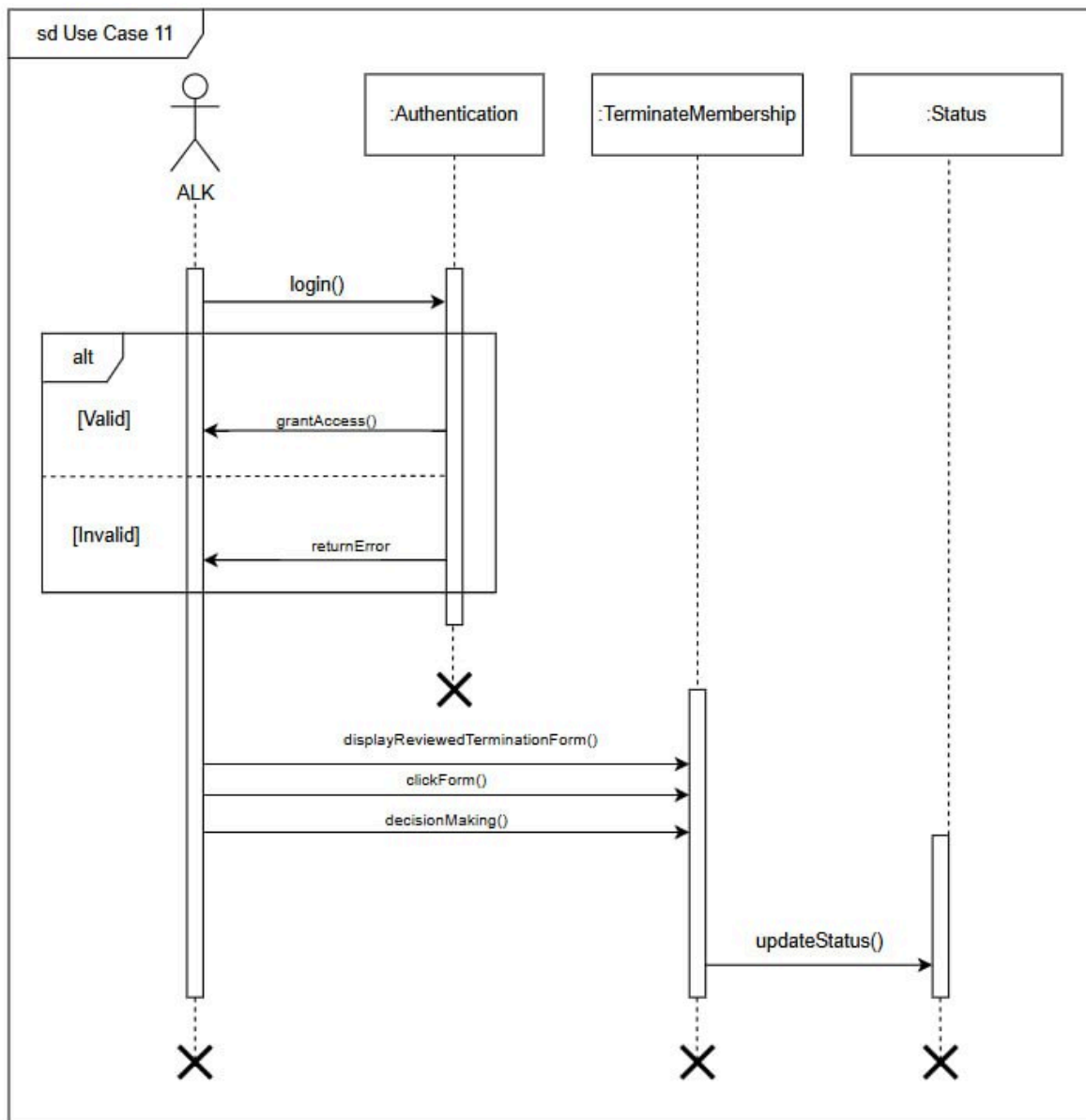


Figure 2.17: Sequence Diagram for Approve Terminate Application

2.3 Launch Phase

Product backlog is a list of tasks and features that need to be done to create a product. It's arranged by priority, with the most important tasks at the top. Each task is

written as a user story, explaining what a user wants from the system. These tasks are picked and worked on during sprints, which are short time periods, usually lasting one to four weeks. During sprints, team members finish tasks, update the system, and make incremental improvements. Each team member is given specific tasks or features to work on, which helps keep things organized and ensures progress is made. Table 2.13 shows the sprint sequence, user story, and which member is responsible for it for Koperasi KADA's Automated System.

Table 2.13: Product Backlog

Sprint	Team members assigned
Sprint #1	Safiya Nursyahadah
Applicant - Login to Apply Membership	
Sprint #2	Safiya Nursyahadah
Admin - Review Applicant's Form	
Sprint #3	Dayang Farah Farzana
ALK - Approve Membership	
Sprint #4	Farra Nurzahin
Applicant - Apply Loan	
Sprint #5	Farra Nurzahin
Applicant - Calculate Loan	
Sprint #6	Dayang Farah Farzana
ALK - Approve Loan Application	
Sprint #7	Nawwarah Auni
Admin - Prepare Statement and Report	
Sprint #8	Nawwarah Auni
Admin - Notify via Email	
Sprint #9	Nawwarah Auni
Admin - Manage Transaction	
Sprint #10	Farra Nurzahin
Applicant - Terminate Membership	
Sprint #11	Dayang Farah Farzana
ALK - Approve Terminate Application	

2.4 User Story Details

This section represents how an applicant, an admin and an ALK would interact with the system performing their respective tasks. It consists of user story description including flow of events, alternative flows, acceptance criteria and exception flow, assuring that every expectation of a user is well handled. It includes uniquely identified user stories with supporting sequence and activity diagrams to elaborate on the process. The structured approach ensures the system is user-centered and in line with the operational objectives of the Koperasi.

2.4.1 US001: User Story Applicant

The user story in Table 2.14 describes the role and actions of the applicant in the koperasi KADA's automated system. The applicant starts by logging into the system with their username and password. If the login is successful, the applicant can access the membership application form and apply for it. Then if membership is approved, applicants can fill out and submit the loan form. If anything is missing while the applicant is filling out the membership and loan form, the system will prompt the applicant to enter the correct information. Applicants also can calculate and display the loan amount along with financial commitments and benefits. Once a member, the applicant also has the option to submit a termination application if they wish to discontinue their membership. The system will guide the applicant through the termination form to ensure all necessary information is met before finalizing the request.

Table 2.14: User Story Description for Applicant

User story: Applicant
ID: US001
User Story Description As an applicant, I want to log into the membership application system, apply for loan and calculate my loan repayment as well as submit a termination application, So that I can apply for membership, request financial support for my needs and to know about my financial commitments and can terminate my membership when necessary.
Flow of events: 1. The Applicant navigates to the login page of the system.

2. The system prompts the Applicant to enter their username and password. 3. The Applicant enters their login details and submits the form. 4. The system verifies the login information. 4.1 If the login details are correct, the system grants access to the Applicant. 4.2 If the login details are incorrect, the system displays an error message. 5. Once logged in, the Applicant can access the membership application form. 6. After approval, applicants can select the “Borang Pembiayaan” option. 7. The system displays the loan application form. 8. Applicant completes the loan application form. 9. Applicant submits the loan application form to the system. 10. Applicants click on “Kalkulator Anggaran Bayaran Balik” and enter input details. 11. The system displays calculated loans based on applicant input. 12. If the Applicant wishes to terminate their membership, they navigate to the “Borang Berhenti” section. 13. The system displays the termination application form. 14. The Applicant completes and submits the termination application form.
Alternative flow 1: 1. If any required fields are empty or invalid, the system will ask the applicant to re-enter the correct or valid information.
Acceptance Criteria Postcondition : 1. The Applicant is successfully logged into the system and gains access to the membership application form. 2. The loan application is successfully submitted and recorded in the system. 3. The system calculates and displays the amount of loan to the member. 4. The termination application is successfully submitted and recorded in the systems. Precondition : 1. The Applicant must be registered in the system with a valid username and password 2. The system must be accessible for login attempts. 3. For loan application, applicants must have an approved membership status. Other conditions : 1. The system must ensure that all login details are encrypted for security purposes. 2. Applicants must meet loan requirements as defined by Koperasi KADA.
Exception flow: 1. If the Applicant enters invalid login details three times, the system locks the account temporarily and prompts Applicant to use the “Forgot Password” option. 2. If the system is unavailable due to maintenance or connection, the applicant can see an error message. 3. If the system is unable to process calculations, an error message will be displayed.

The sequence diagram in Figure 2.18 illustrates the interactions between the Applicant, Authentication, Membership, Loan and Terminate Membership components in handling use cases 1, 4, 5 and 10. It begins with the Applicant attempting to log in through the login process. Depending on the outcome, either the applicant page is displayed for successful login or an error message for failure. Upon accessing the membership section, the admin can interact with the Membership component by accessing and submitting the membership form, followed by receiving membership status. For loan management, if the user is a member, the system facilitates actions like displaying the loan form, submitting it, calculating loan details, and presenting the results. If the user is not a member, an error message is shown. For termination, if a member wishes to terminate their membership, they can submit a termination request.

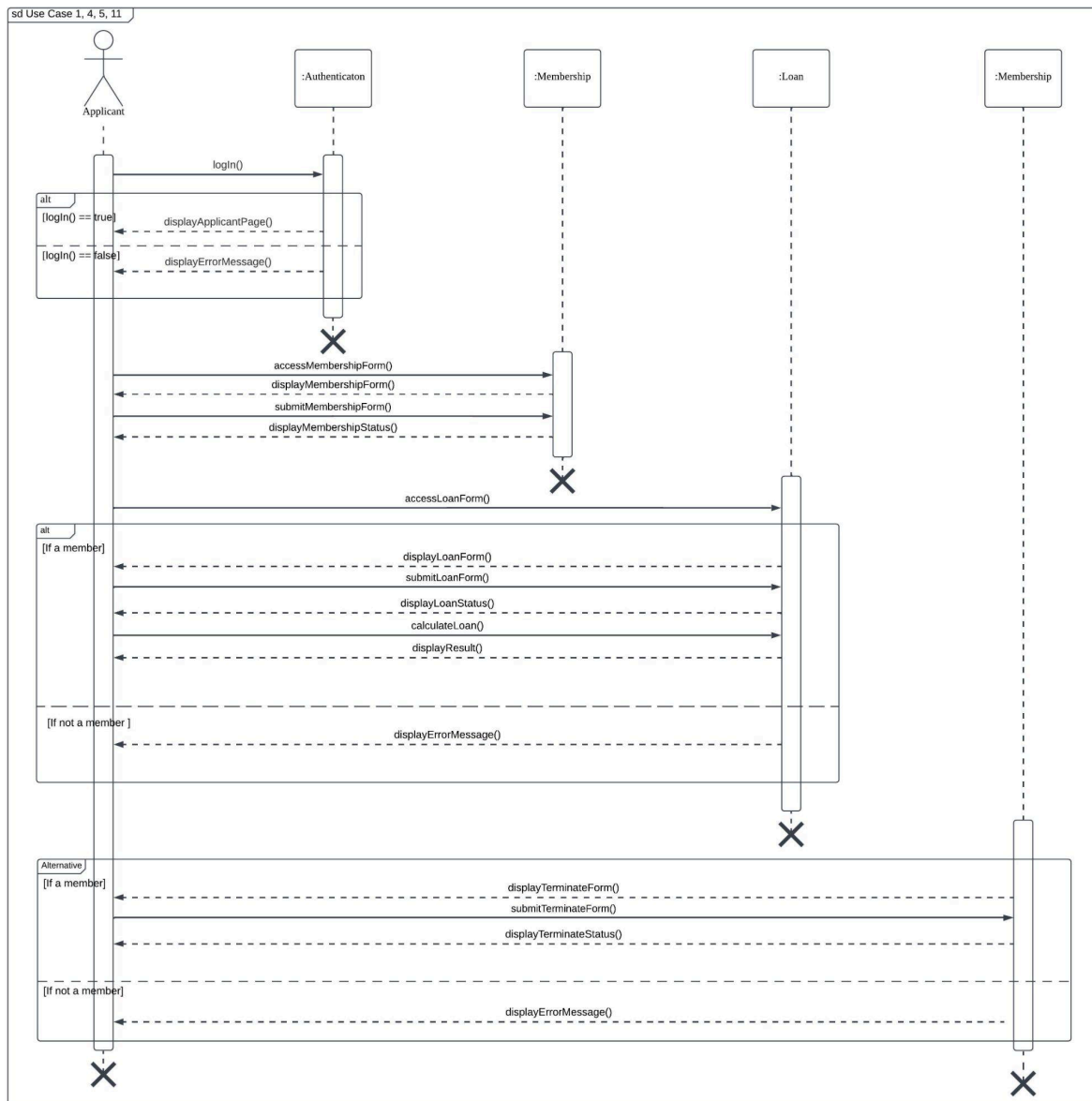


Figure 2.18: Sequence Diagram for Applicant

The activity diagram in Figure 2.19 explains the steps an applicant follows when using the system for membership and loan applications. It starts with the applicant logging in. If the applicant is not authorized, an error message is displayed, otherwise, they are redirected to the applicant page. For membership, the applicant selects "Apply Member," fills in the membership form, and submits it. If the form is incomplete, they need to re-enter the missing details. Once complete, the system displays the membership status. For loan applications, the applicant selects "Apply Loan," fills in the loan form, and submits it. If the form is incomplete, they re-enter the missing details. Afterward, the applicant can use the loan calculator to input loan details. If the input is valid, the system calculates and displays

the calculated loan and dividend. For termination, the applicant selects “Borang Berhenti”, fills in the termination form and submits it.

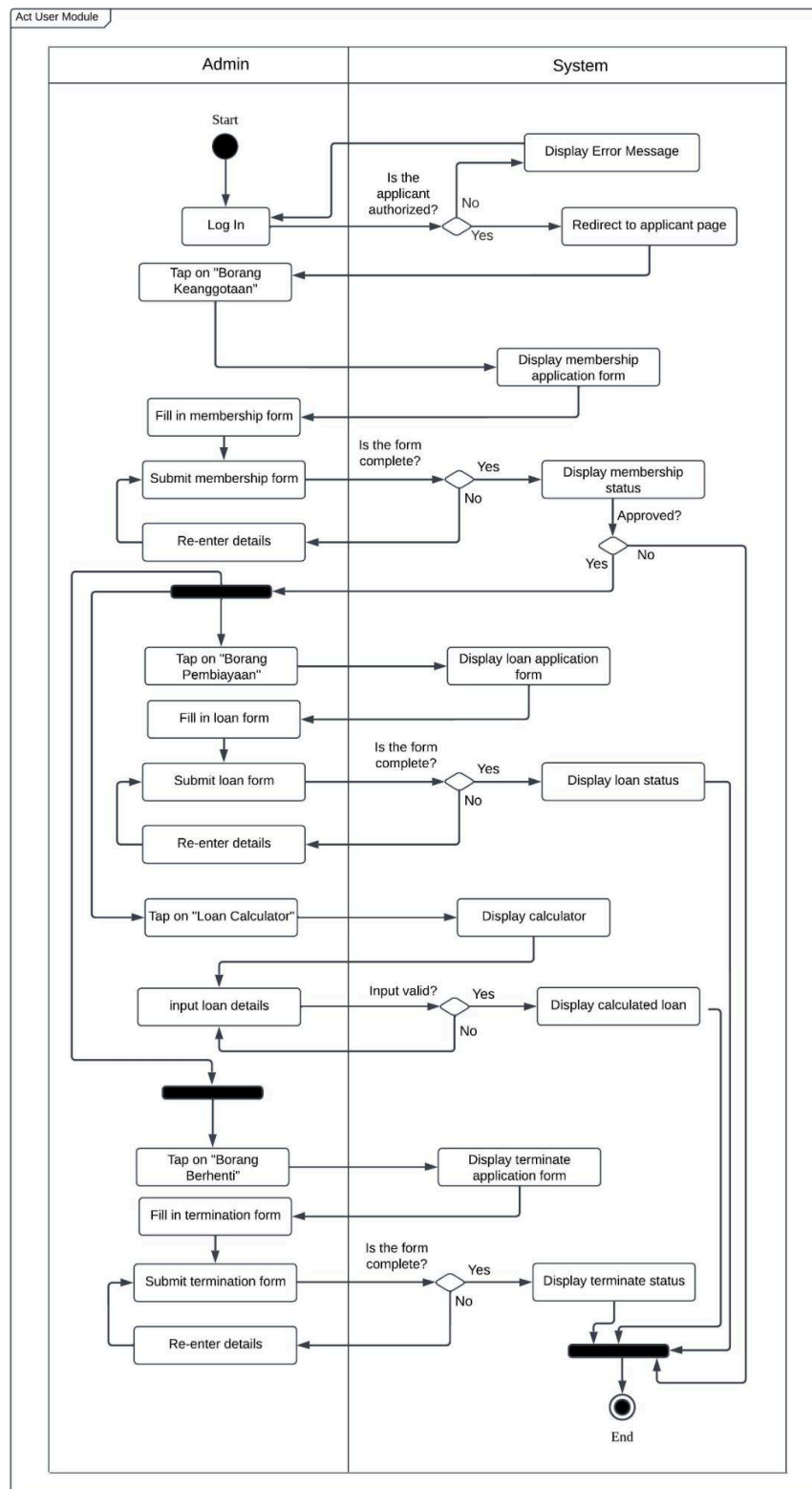


Figure 2.19: Activity Diagram for Applicant

2.4.2 US002: User Story Administrator

Table 2.15 is a user story description for an admin managing the system to handle tasks such as reviewing membership and loan application, generating statements and sending automated notifications. The admin starts by logging into the system where their login details are verified. If the login details are correct, the admin is granted access to the admin page. If incorrect, an error message is displayed. The admin can select from various options at the admin page once logged in. If the admin chooses to review applications, they can click to either the 'Membership List' or 'Loan Application List'. If the admin selects the 'Approve Member by ALK', they can generate an online statement. This process includes creating the statement by clicking 'Generate Statement' and then sending an automated notification to the applicant by clicking 'Push email to applicant'.

Table 2.15: User Story Description for Administrator

User story: Administrator
ID: US002
User Story Description As an Admin, I want to review and process membership applications, loan applications, and membership termination requests, as well as manage transactions, prepare statements, and send notifications, So that I can ensure the accuracy, completeness, and proper handling of financial, membership and loan related processes.
Flow of events: 1. The Admin logs into the system. 2. The Admin navigates to the section for reviewing membership applications, loan applications, or termination requests. 3. The system displays a list of pending applications. 4. The Admin selects an application to review. 5. The system displays the application details. 6. The Admin verifies the provided information. 7. If the application is complete and valid, 7.1. it is marked as approved. 8. If the application is incomplete or contains errors, 8.1. it is flagged for follow-up. 9. The Admin prepares statements and reports based on applicant data. 10. The Admin generates and sends notification emails regarding application status and financial statements. 11. The Admin manages and records financial transactions, whether monthly or irregular.
Acceptance Criteria Postcondition : 1. The application review process is completed, and the applicant's details are verified. 2. Statements and reports are successfully generated and saved.

3. Email notifications are sent to applicants. 4. Financial transactions are recorded accurately.
Precondition : 1. The Admin must have valid login details to access the system. 2. Applicants must have submitted their membership and loan applications forms for review. 3. Required data for the statement and report must be accessible within the system. 4. The system must have a list of recipients with valid email addresses. 5. The statement or report has been prepared in the system. Other conditions : 1. The system must allow the Admin to view all details of the selected application form. 2. The system must ensure that only pending applications are displayed for review. 3. The system must have sufficient access control to ensure only authorized admins can generate reports. 4. The admin must be able to view and select the specific applicant for whom the statement or report is to be generated.
Exception flow: 1. If an application form is incomplete or contains errors, the Admin flags it for further action. 2. If the system fails to generate the statement or report due to system errors, the admin will be notified with an error message. 3. If the system fails to generate a notification email, it will notify the admin of the failure and allow the admin to retry sending the email.

Figure 2.20 shows the sequence of steps when an admin uses the system, from logging in to managing applications, transactions, and notifications. First, the admin logs in, and the system checks if the details are correct. If valid, the admin can view pending applications, review them, whether the application is completed or uncompleted. After that, the admin can process financial transactions, update records, and generate reports. If everything is successful, a confirmation message appears. The system also allows sending notifications to applicants. If something goes wrong, like a failed transaction or error in generating a statement, an error message is shown to inform the admin.

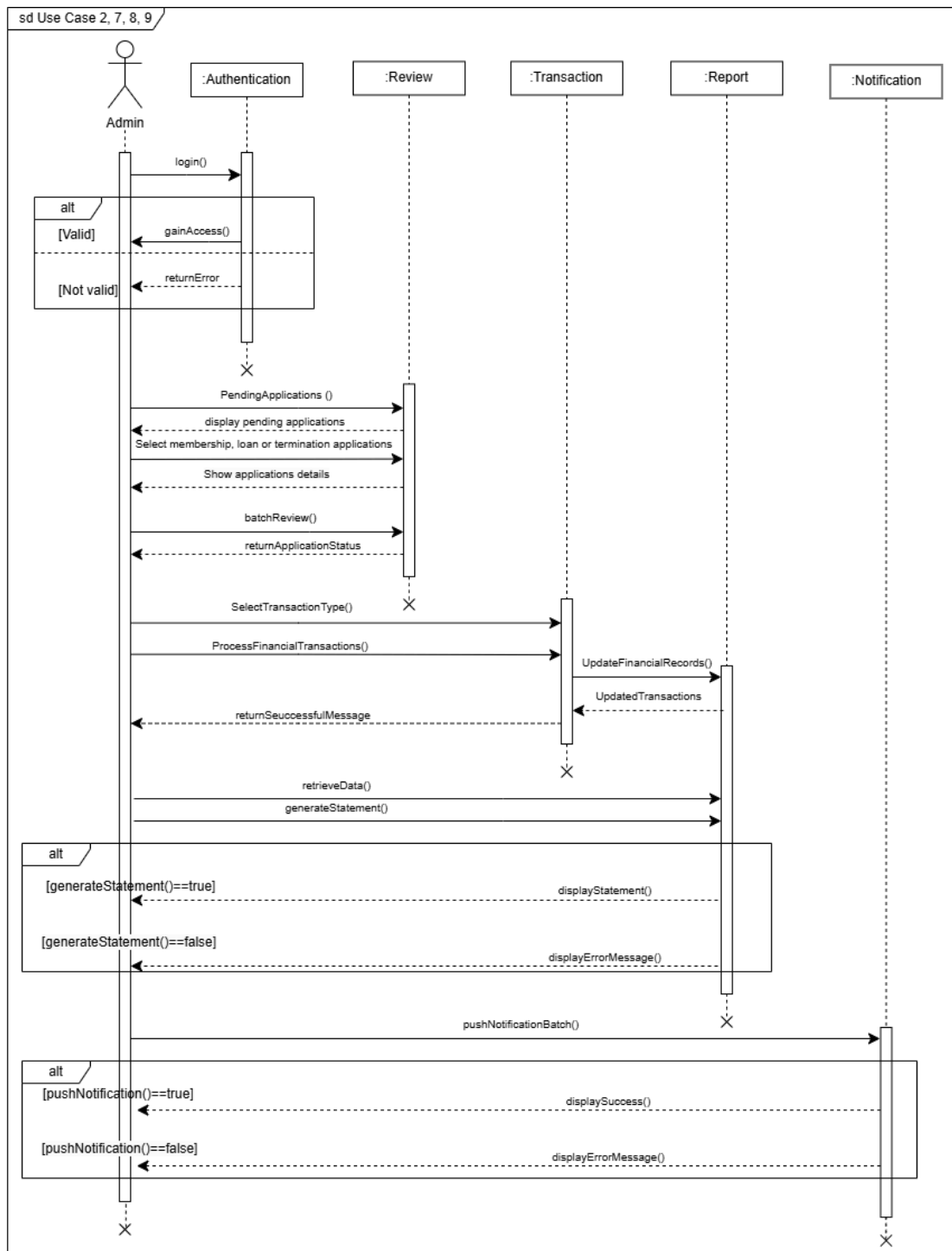


Figure 2.20: Sequence Diagram for Administrators

Figure 2.21 is an activity diagram that presents the flow of how an admin manages user applications and financial transactions. The process starts with the admin logging in, where the system verifies the credentials. After logging in, the admin can review membership or loan applications, generate statements, or send notifications to applicants.

Applications are either marked as complete or incomplete based on the provided details during the review. If an application is approved, the admin can proceed with financial transactions, and the system updates the records accordingly. Once the transactions are completed, the admin generates statements, and the system prepares a confirmation for review before sending them to approved members. Notifications are then sent to applicants, informing them about their application status or financial updates. If any step fails, the system displays an error message to guide the admin. The diagram ensures a clear and organized process for handling all admin tasks efficiently.

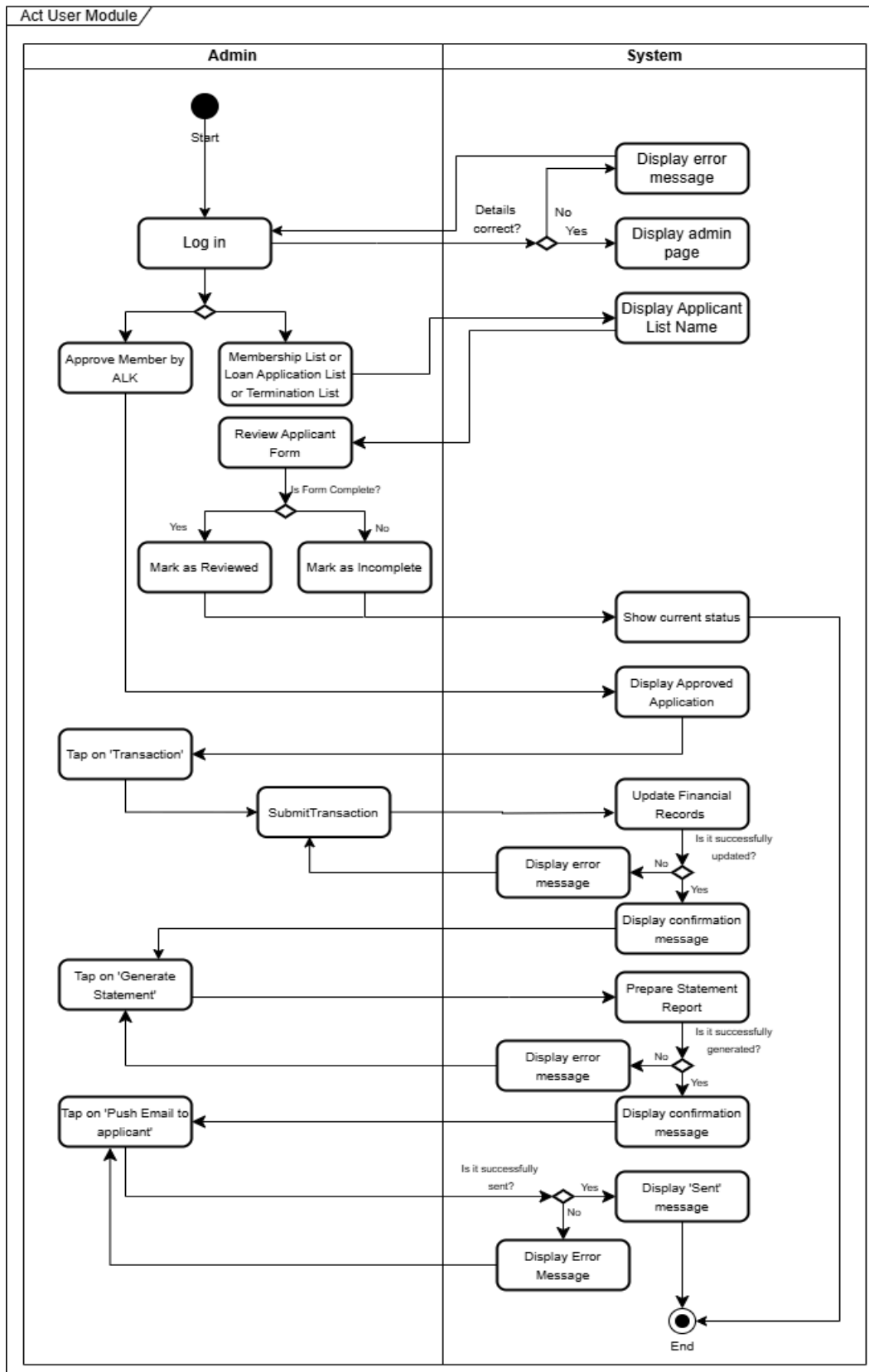


Figure 2.21: Activity Diagram for Administrators

2.4.3 US003: User Story Ahli Lembaga Koperasi (ALK)

Table 2.16 shows a process of ALK in the membership, loan and membership termination approval. In this system, ALK will log in, choose a membership application, loan application, or membership termination application to review the pending applications for approval or rejection. An incomplete application marked by Admin is directly rejected by the system, and applications which need further discussion are marked for meeting actions. Once the decisions are made, the system updates and records the statuses. The Table also states the precondition that Admin has to go through the applications beforehand.

Table 2.16: User Story Description for ALK

User story: ALK
ID: US003
User Story Description As an Ahli Lembaga Koperasi (ALK), I want to approve applicants' membership and loan application, So that I can ensure that I can ensure only eligible individuals can become membership and receive loan from the company
Flow of events: 1. ALK logs into the system 2. ALK choose to evaluate membership, loan application or terminate membership application 2.1. If ALK chose membership 2.1.1. ALK accesses the pending list of membership applications 2.1.1.1 If ALK choose batch processing 2.1.1.2 ALK tick checkbox 2.1.1.3 ALK approve or reject selected applications 2.1.2. ALK selects any application from the pending list 2.1.3. ALK evaluate the application 2.1.4. ALK decides to approve or reject the membership application 2.1.5. ALK chooses next membership application 2.1.6. ALK log out of the system 2.2. If ALK chose loan 2.2.1. ALK accesses the pending list of loan applications 2.2.1.1 If ALK choose batch processing

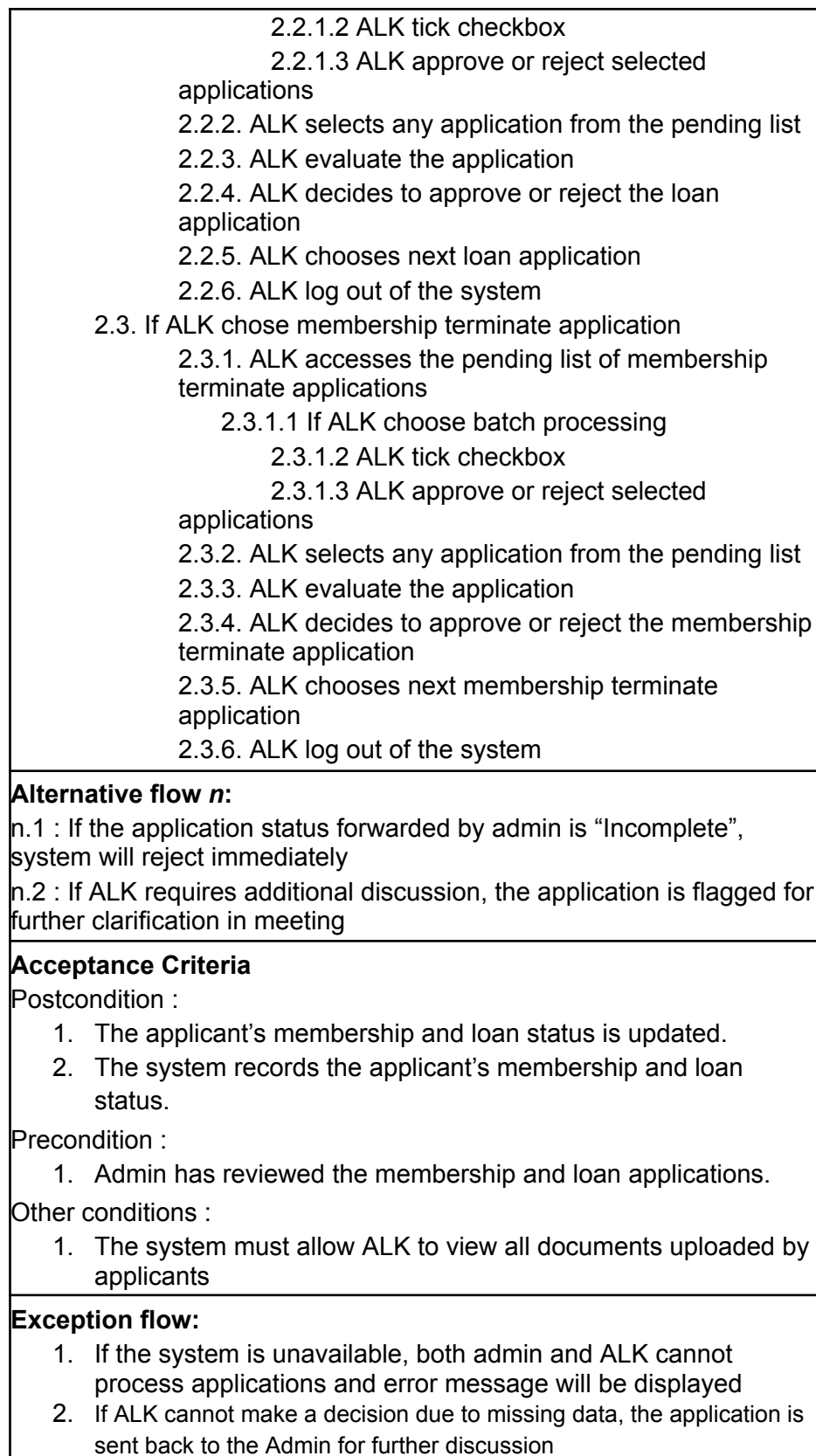


Figure 2.22 is a sequence diagram that shows the interaction between ALK and the system components. The process starts when ALK logs into the system. The system checks

the validity and grants access or returns an error. After logging in, ALK can either access the Membership, Loan and Membership Termination Module. In the Membership module, ALK starts this process by selecting the Membership List. The system opens a list of pending membership applications. ALK opens an application by selecting, makes a decision to approve or reject and the system updates the membership status. ALK also can utilize batch processing by selecting intended membership applications to be approved or rejected simultaneously. In the loan module, ALK selects Loan List, then reviews a pending loan application and may want to compare the application details with loan terms. After that, the system provides the comparison results. ALK then approves or rejects the loan application, then the system will update the status of the loan application. ALK is also allowed to use batch processing for the approval process. In the Membership Termination Module, ALK starts by choosing the Termination Application List. The system displays a list of pending membership applications. ALK opens an application by selecting, makes a decision to approve or reject and the system updates the membership status. ALK also can use batch processing by selecting intended membership termination applications to be approved or rejected simultaneously.

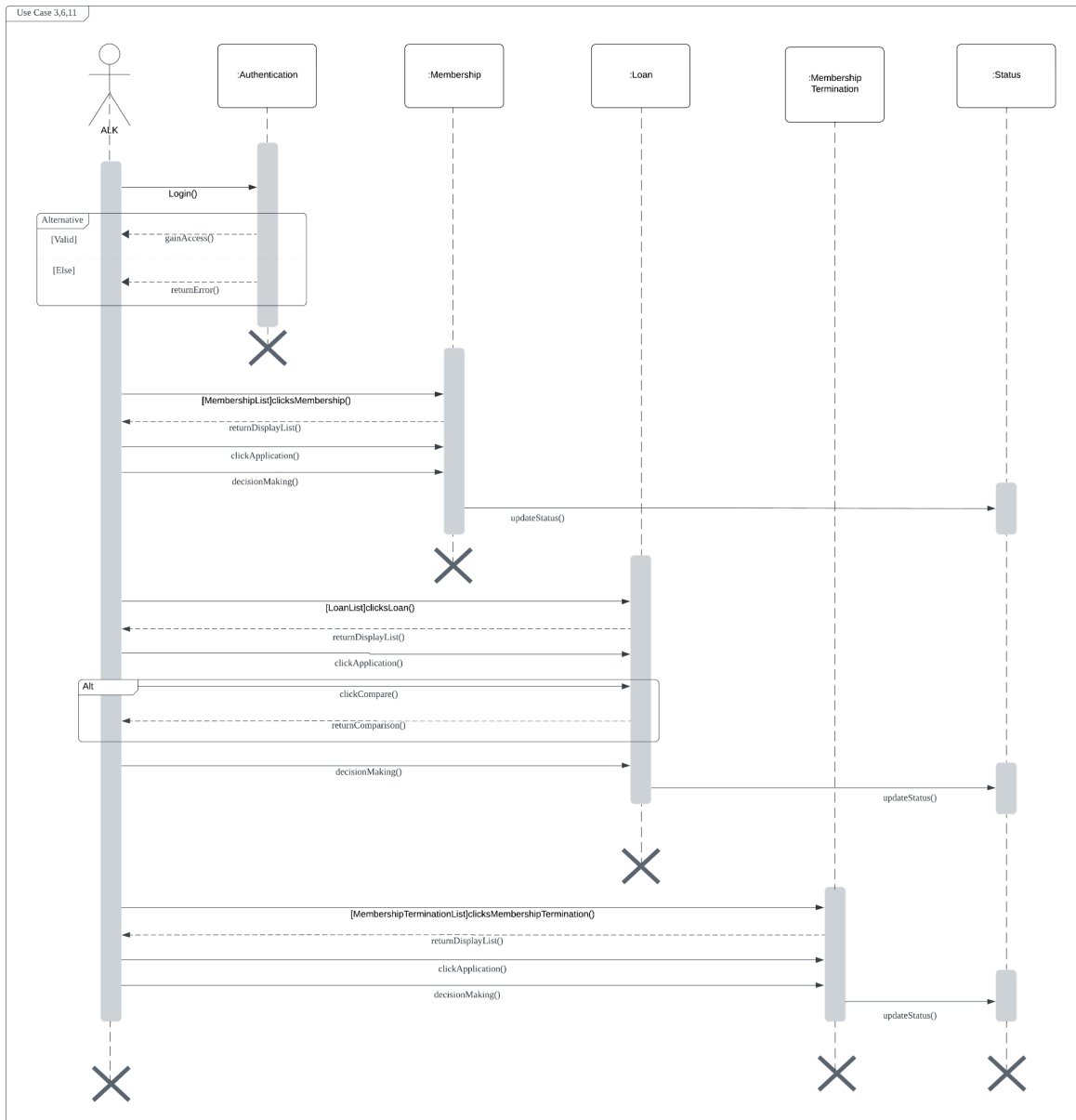


Figure 2.22: Sequence Diagram for ALK

Figure 2.23 shows the activity diagram of the workflow between ALK and the system for managing membership and loan applications. ALK logs in, and upon successful access, selects to review membership, loan or membership termination applications. In the case of membership, ALK will assess pending applications, approve or reject them and the system will update the status. For loans, ALK may directly assess or compare to evaluate. The system will show the result of comparison to assist ALK in decision making. Approved or rejected statuses are recorded, thus ensuring systematic and informed application

management. For membership terminations, ALK will assess pending applications, approve or reject the applications and the status will be updated into the system.

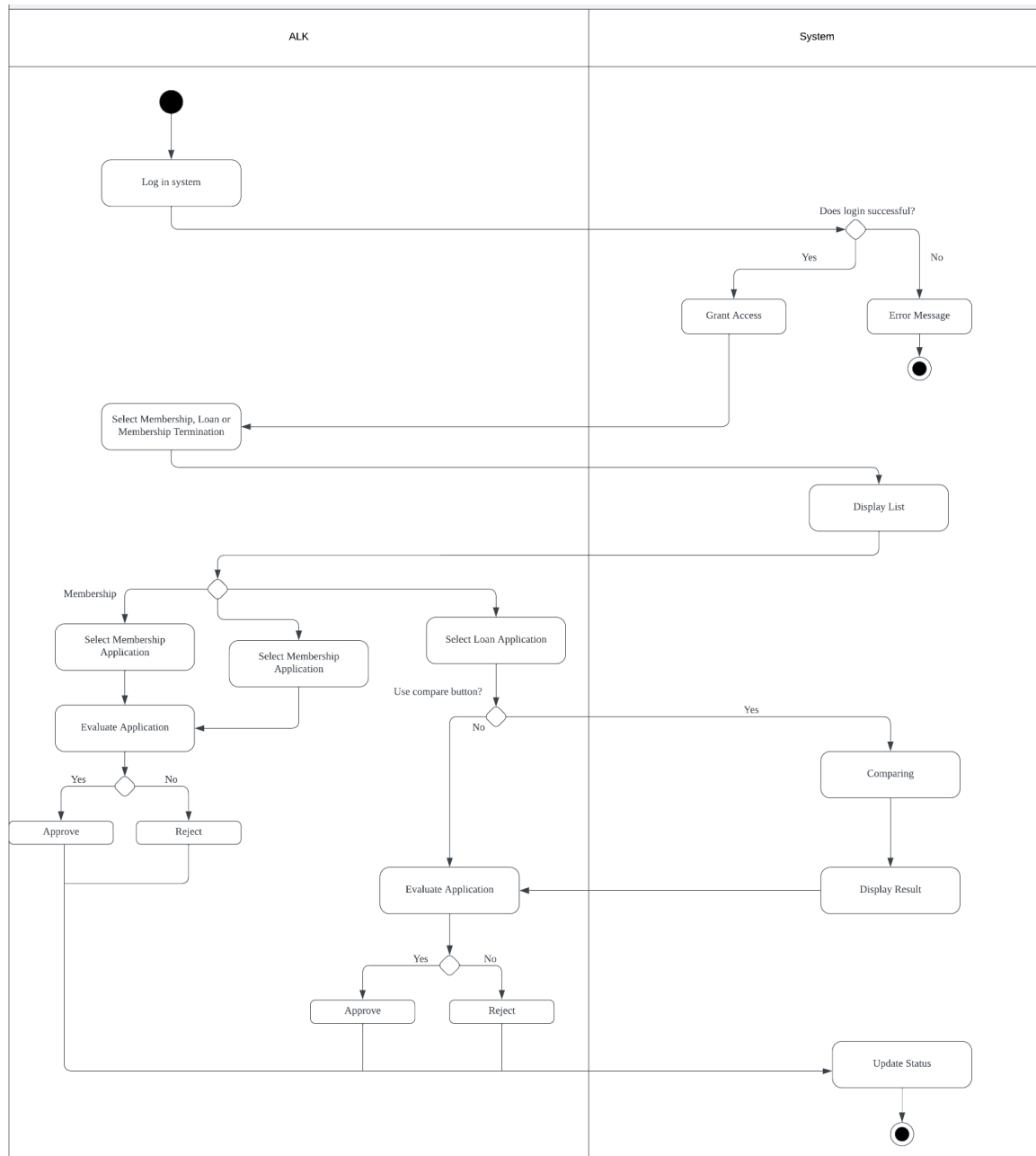


Figure 2.23: Activity Diagram for ALK

2.5 Performance and Other Requirements

Non-functional requirements for this project system define how the system should work and focus on that rather than show its specific functionalities. It is essential to have

non-functional requirements to make sure the system fulfills quality standards and can work in various environments. These requirements are classified into Software System Attributes, Performance and Other Requirements according to the ISO/IEC/IEEE 29148 standard.

Performance requirements specify how quickly the system can handle data and react to user requests. These are requirements that belong to the system's performance under different conditions such as:

- Response time: The response time for any user request should not exceed 2 seconds under normal operating conditions.
- Throughput: The system should handle a significant number of concurrent users without performance degradation to ensure timely processing of membership and loan applications.
- Capacity: The system must store a large number of member records and loan applications while maintaining optimal performance.
- Availability: The system should remain accessible and functional even under interruptions or heavy loads.

Software System Attributes define the overall quality and behavior of the software. These attributes include:

- Usability: The system's interface should enable new users to complete tasks without requiring a lot of training.
- Reliability: The system must secure data consistency and correctness, with a monthly downtime tolerance of no more than 0.5%.
- Maintainability: The system should enable updates such as bug fixes by laying objectives to finish major updates in a week and minor updates in 1 to 2 days.
- Portability: The system's software needs to work with a variety of hardware and software operating systems, including different Windows and macOS versions.
- Safety: The system includes specific safety requirements to mitigate potential loss, damage, or harm associated with its use, particularly in handling sensitive financial and personal data.
- Security: The system will require multi-factor authentication (MFA) to help prevent unauthorized access.

2.6 Design Constraints

The design constraints for the system focus on ensuring it works efficiently with standard computers and devices, while maintaining security for user data. It should be

compatible with common hardware and across different operating systems and devices. Additionally, the system should support multiple users and provide quick access to information.

- **Hardware Constraints:** The system must be designed to operate on standard desktop computers with a minimum of 8GB of RAM and a dual-core processor to ensure efficient performance. It should also be compatible with existing printers and scanners to make document management easy. The system will need to have the proper USB or network ports to connect with these devices without any issues.
- **Security Constraints:** The system must follow data privacy rules and have security measures to protect user information. This includes secure login, encryption of personal data, and access controls to stop unauthorized users from accessing information. Additionally, regular security updates should be applied to maintain system integrity and prevent potential threats.
- **Compatibility Constraints:** The system should be compatible with the hardware and software that are mostly used by applicants, admin and ALK. It has to be compatible with different operating systems, browsers and devices in order to be accessible to everyone. The system should be optimized for both desktop and mobile devices to cater to different user needs.
- **Performance Constraints:** The system must be able to handle at least 500 simultaneous users without a noticeable decrease in performance. Data retrieval and report generation should be completed in under 5 seconds for typical queries to provide fast access to information. Additionally, the system must maintain 99.9% uptime, ensuring that it is always available for daily operations without interruptions.